

BFS Traversal

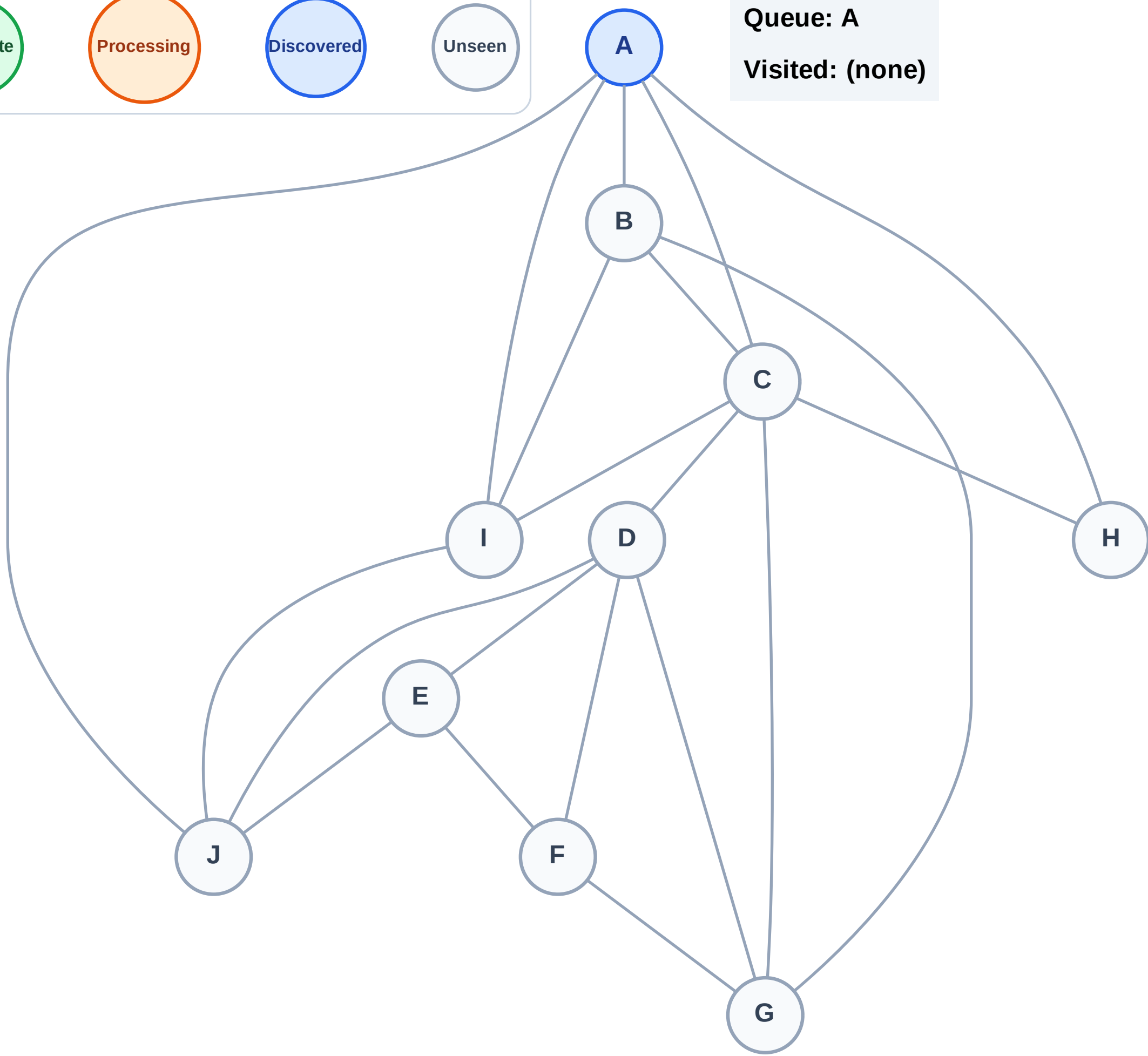
Step 1: Start at A

Legend



Queue: A

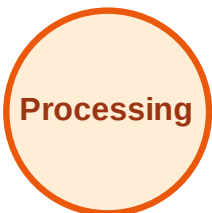
Visited: (none)



BFS Traversal

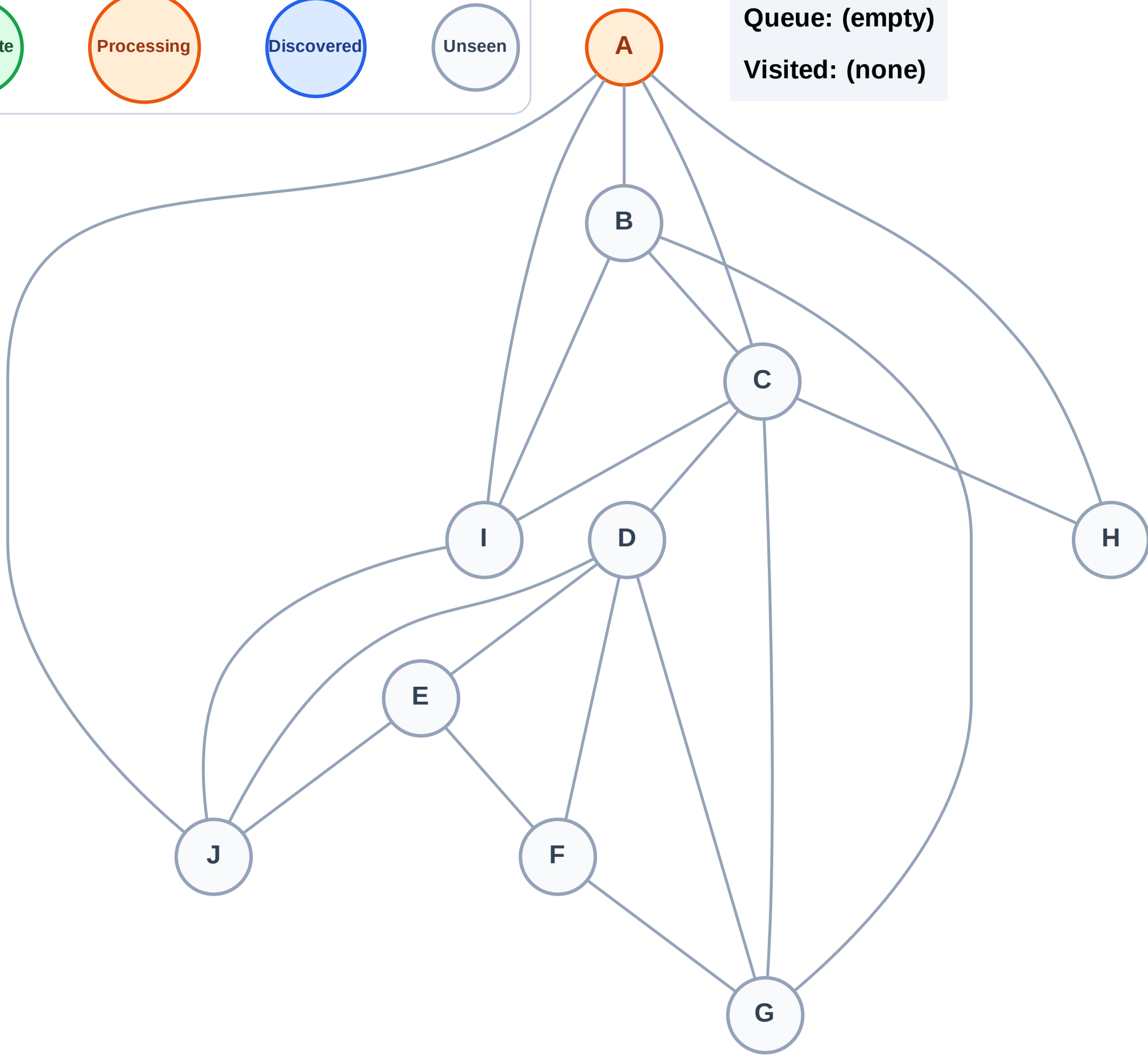
Step 2: Process node A

Legend



Queue: (empty)

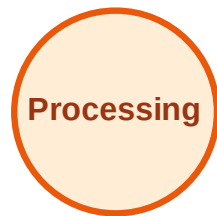
Visited: (none)



BFS Traversal

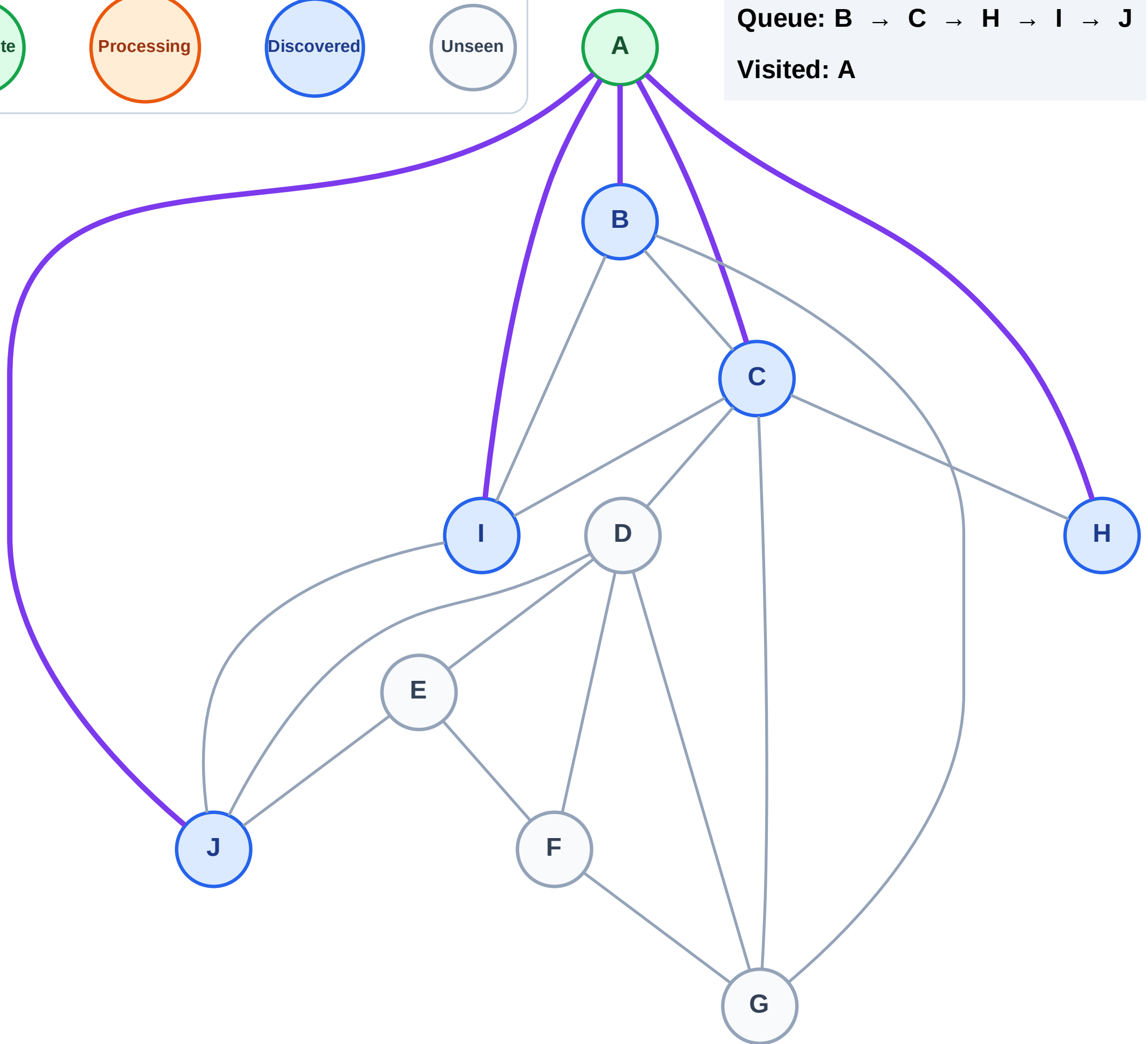
Step 3: Discover B, C, H, I, J from A

Legend



Queue: B → C → H → I → J

Visited: A



BFS Traversal

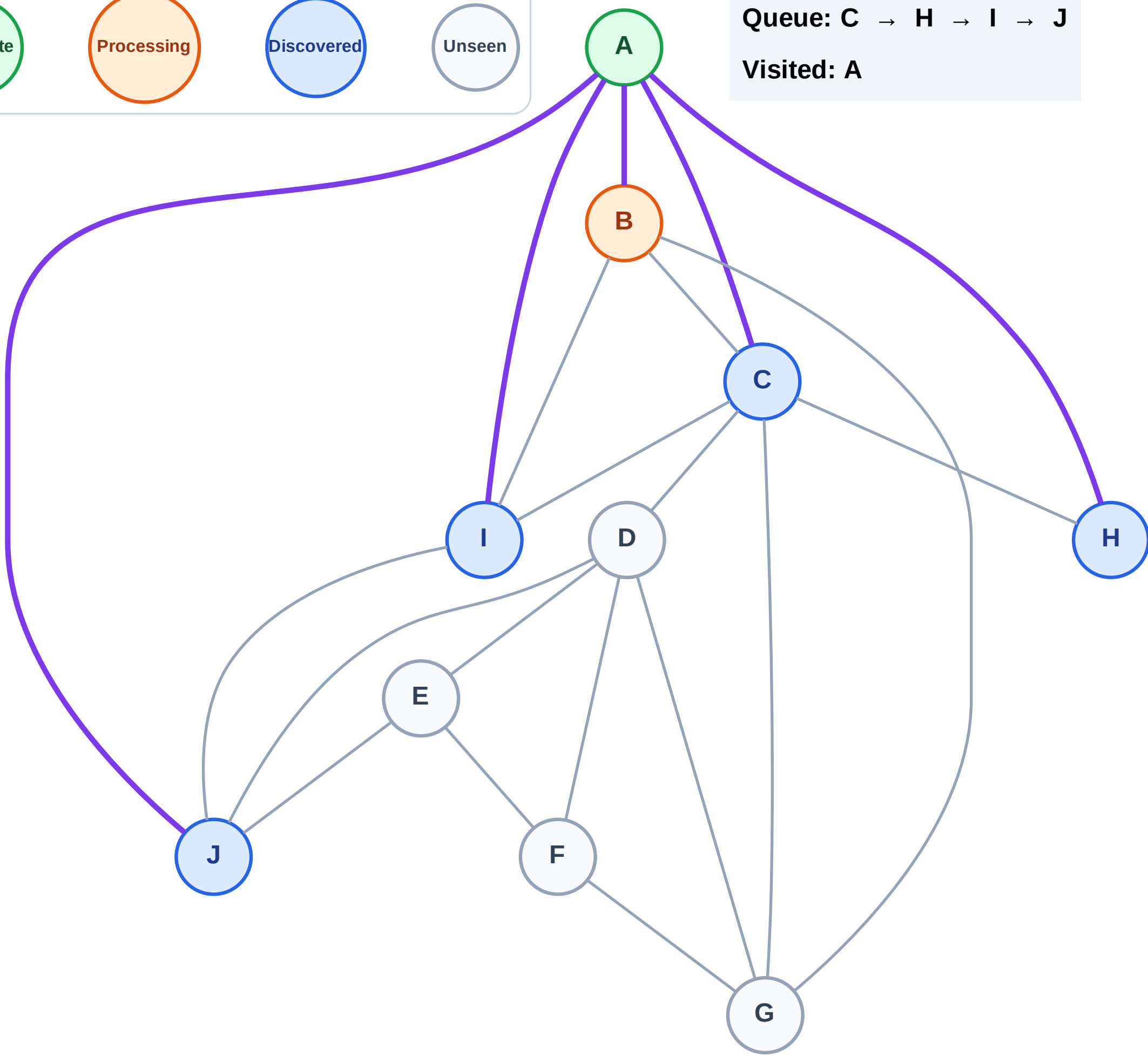
Step 4: Process node B

Legend



Queue: C → H → I → J

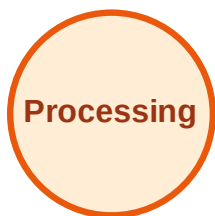
Visited: A



BFS Traversal

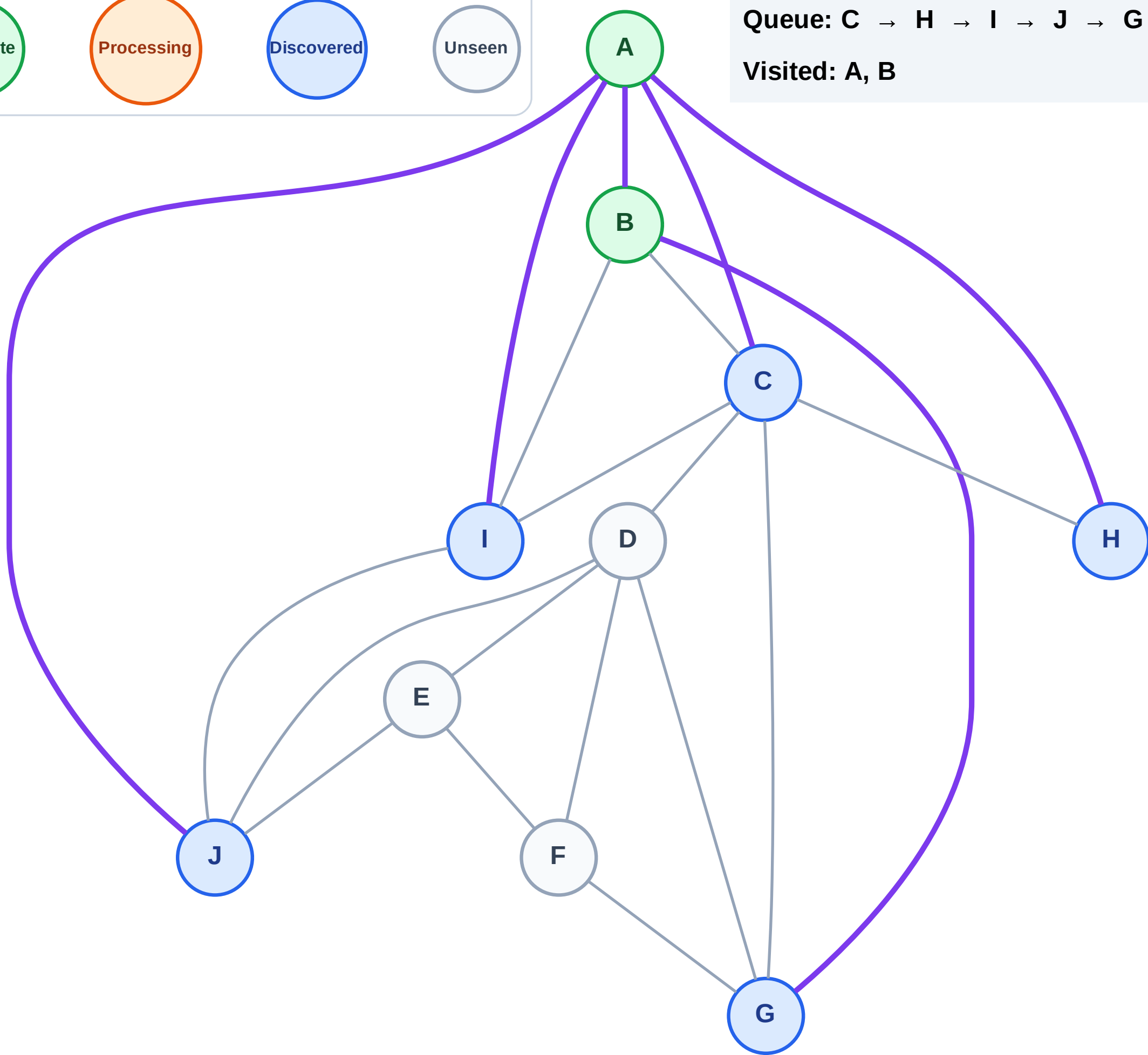
Step 5: Discover G from B

Legend



Queue: C → H → I → J → G

Visited: A, B



BFS Traversal

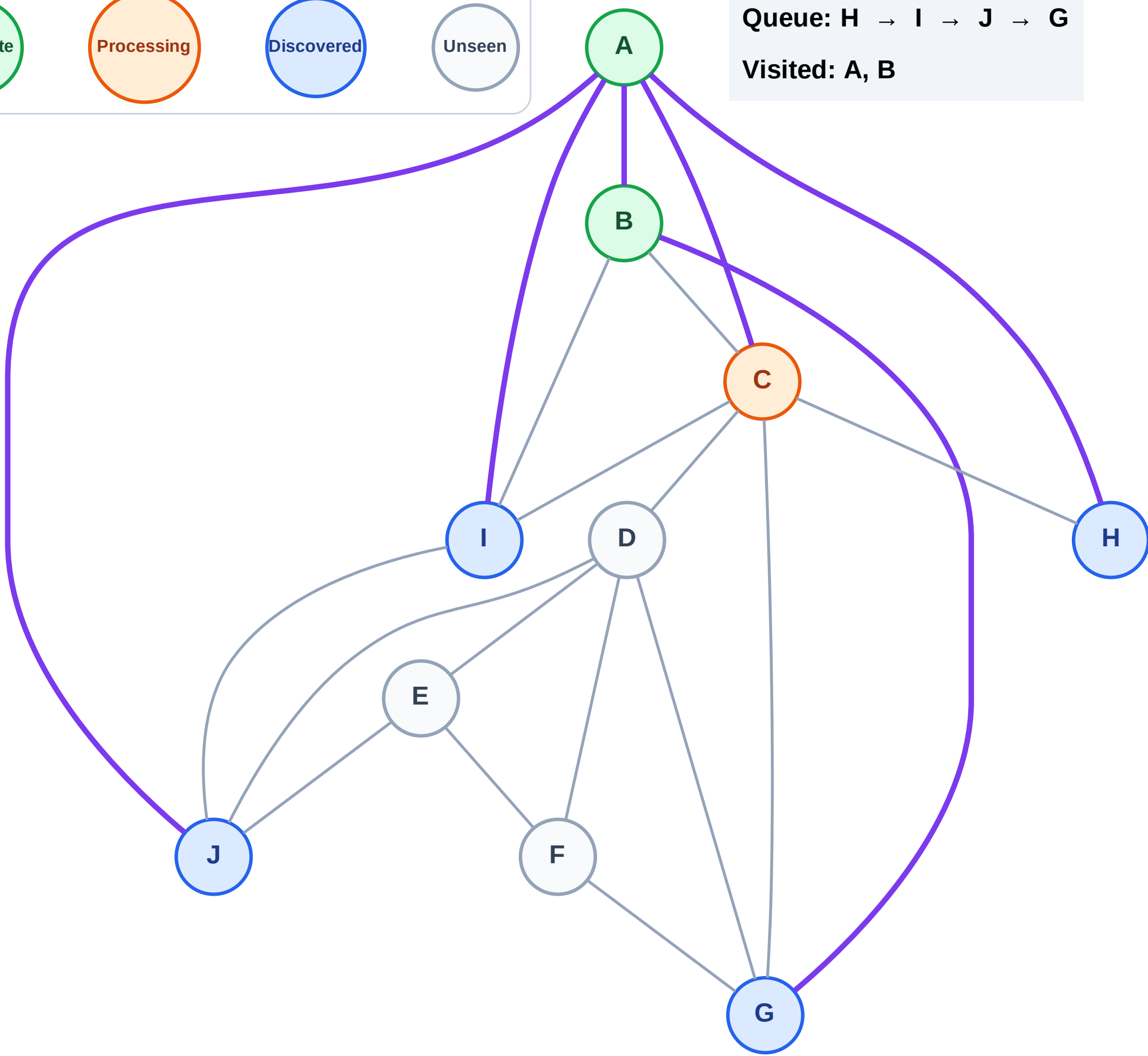
Step 6: Process node C

Legend



Queue: H → I → J → G

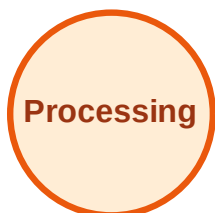
Visited: A, B



BFS Traversal

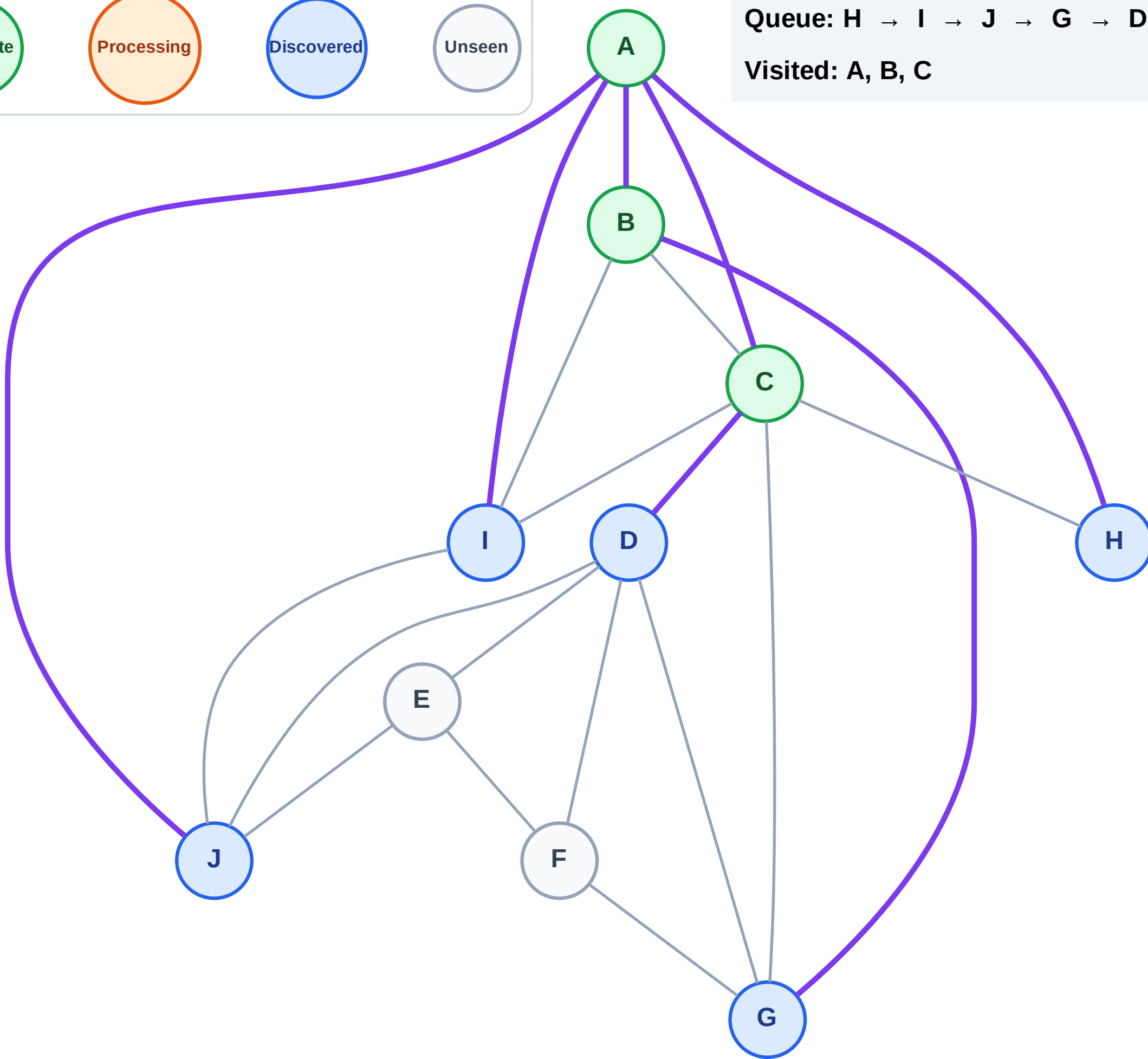
Step 7: Discover D from C

Legend



Queue: H → I → J → G → D

Visited: A, B, C



BFS Traversal

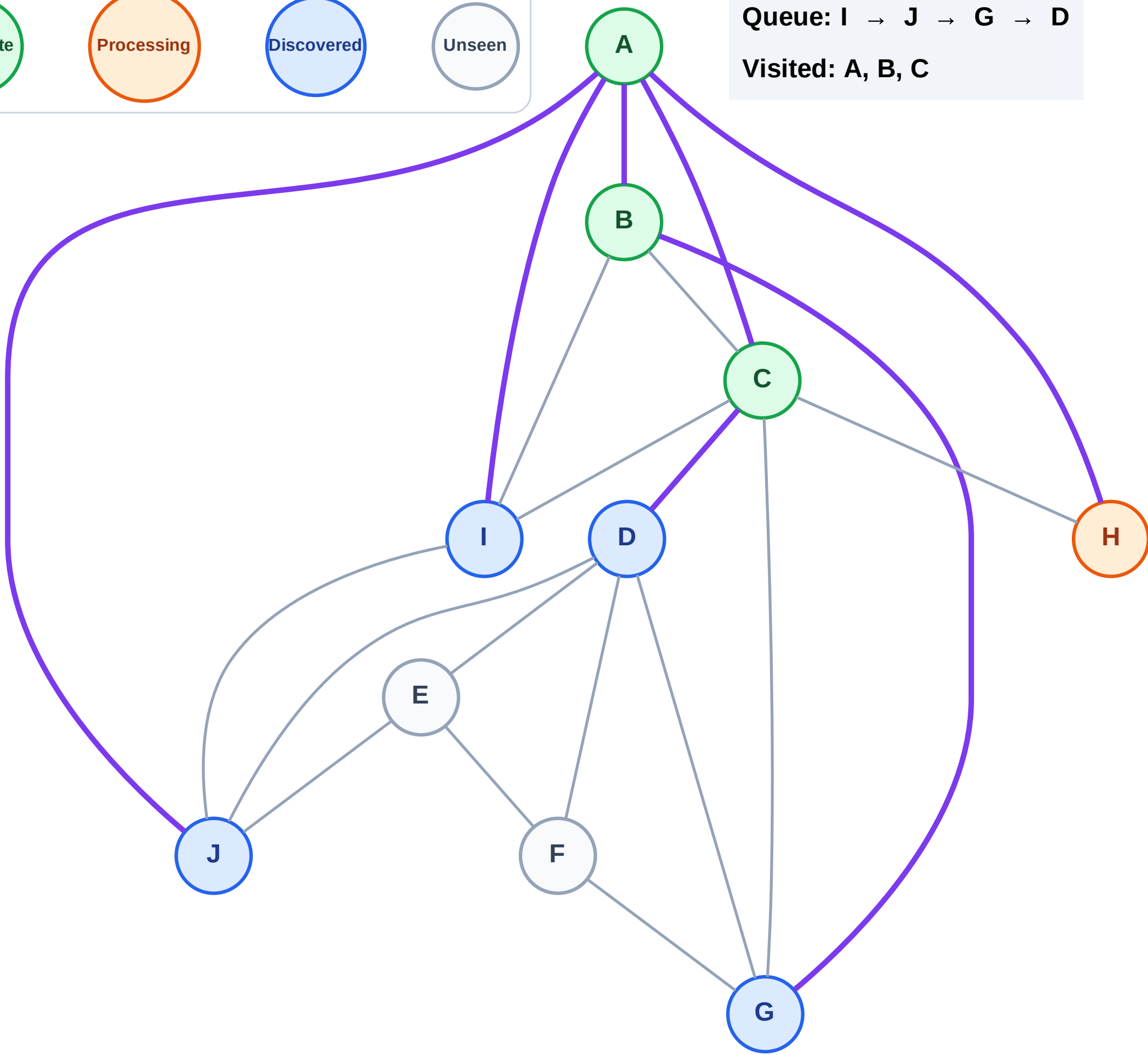
Step 8: Process node H

Legend



Queue: I → J → G → D

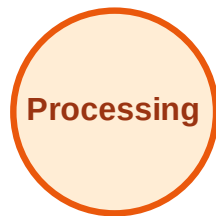
Visited: A, B, C



BFS Traversal

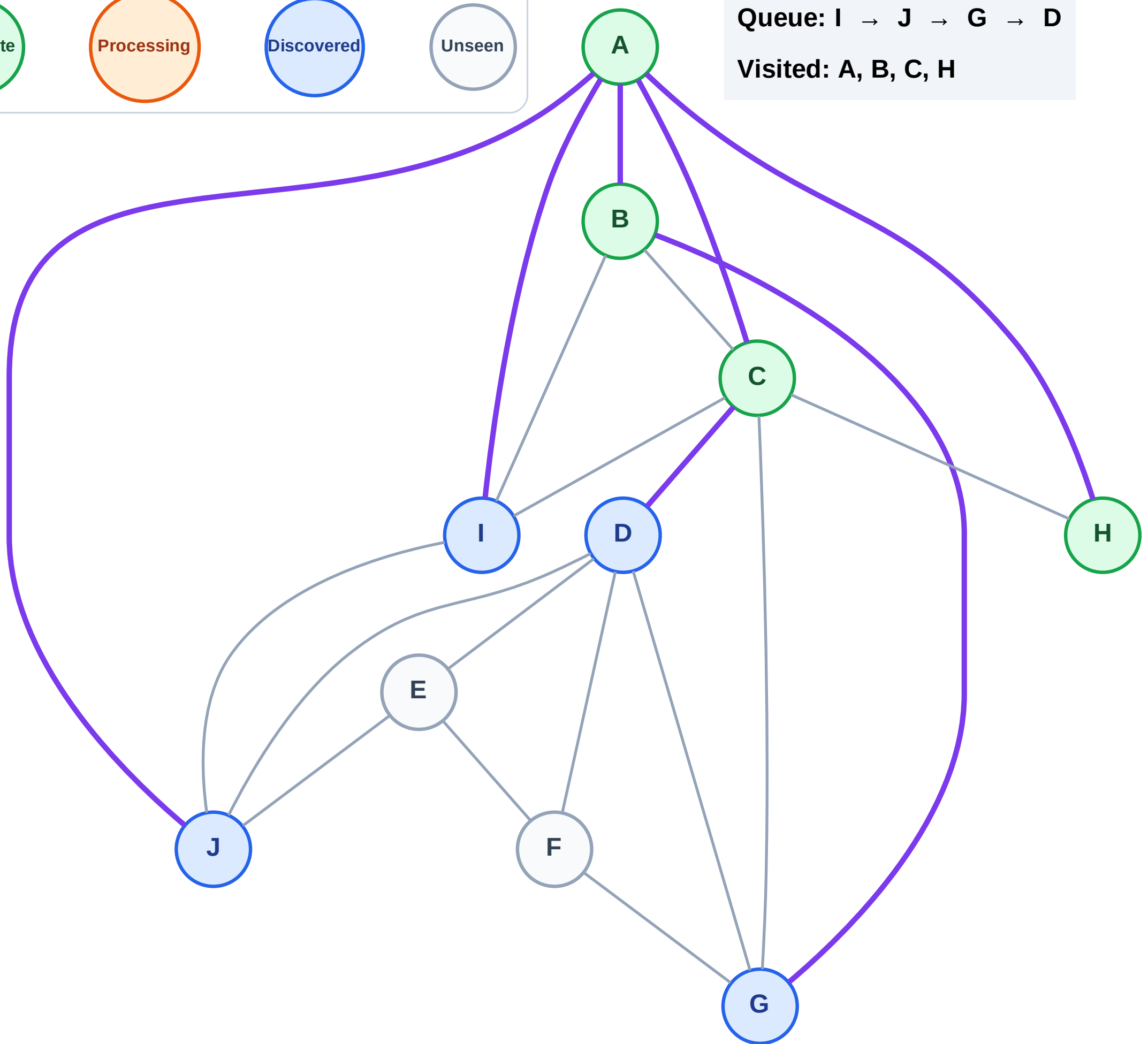
Step 9: No new neighbors from H

Legend



Queue: I → J → G → D

Visited: A, B, C, H



BFS Traversal

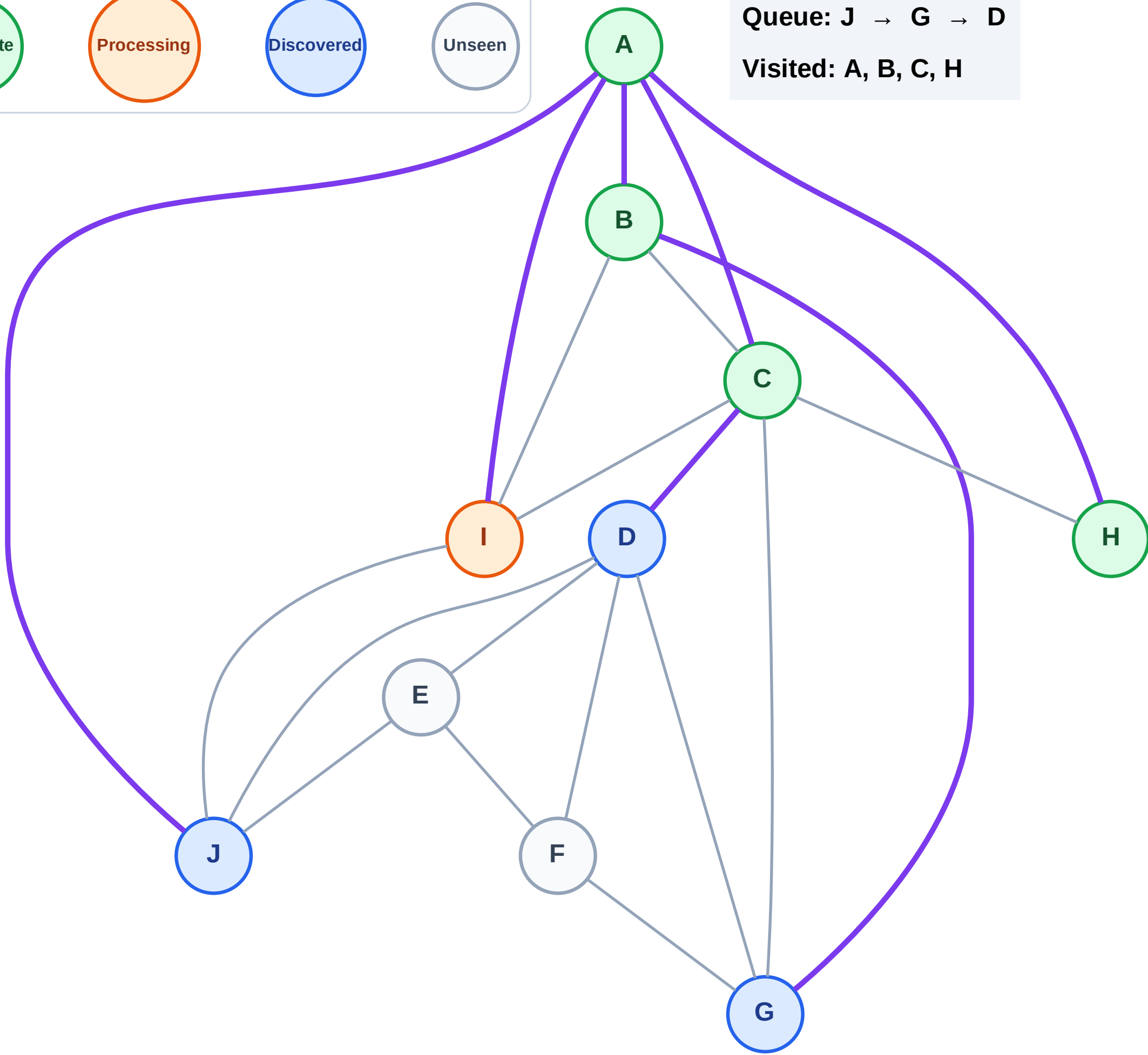
Step 10: Process node I

Legend



Queue: J → G → D

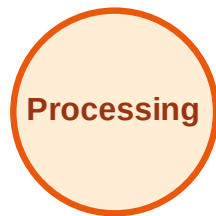
Visited: A, B, C, H



BFS Traversal

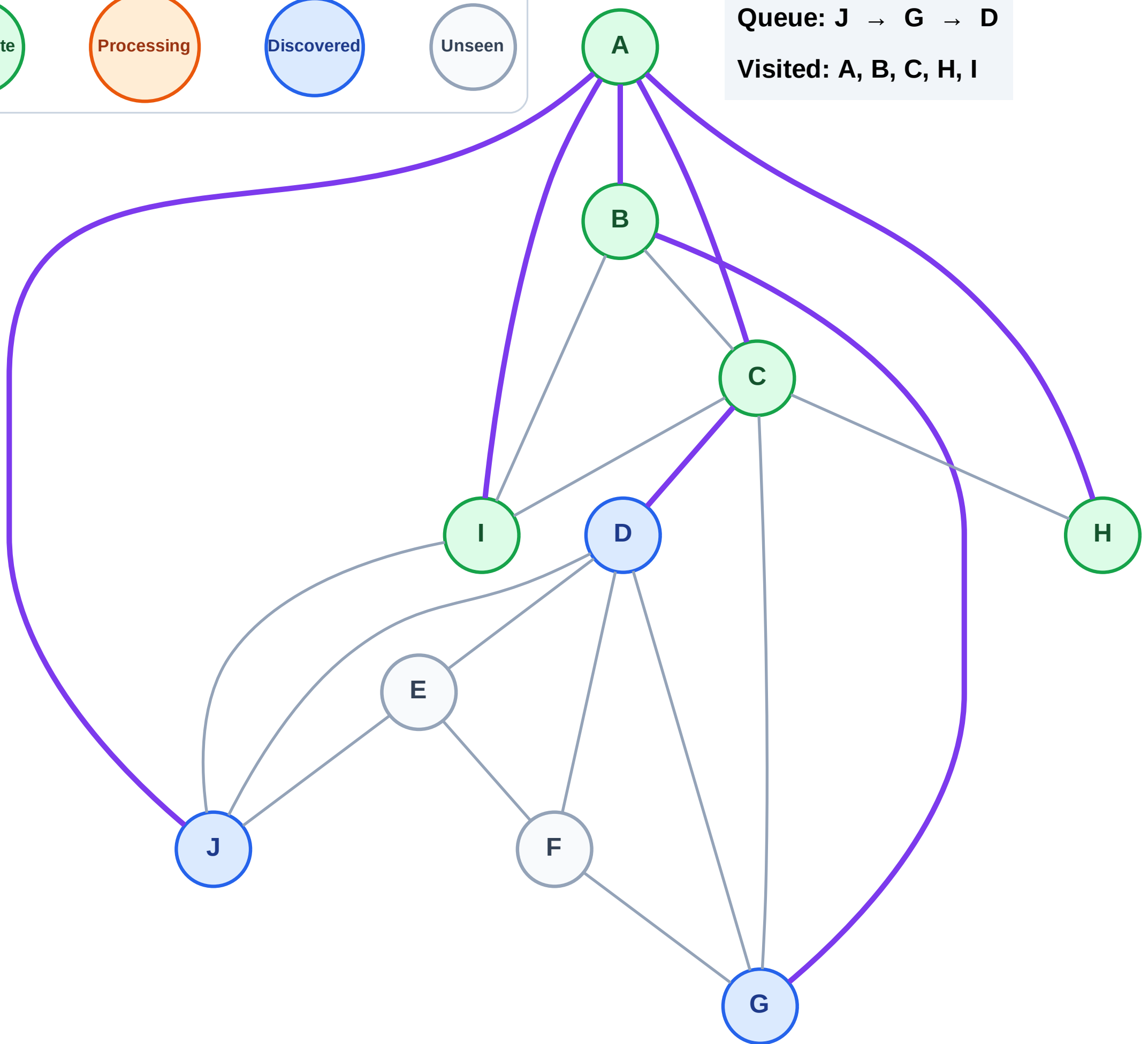
Step 11: No new neighbors from I

Legend



Queue: J → G → D

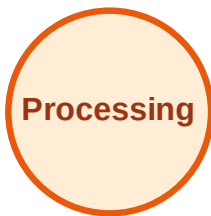
Visited: A, B, C, H, I



BFS Traversal

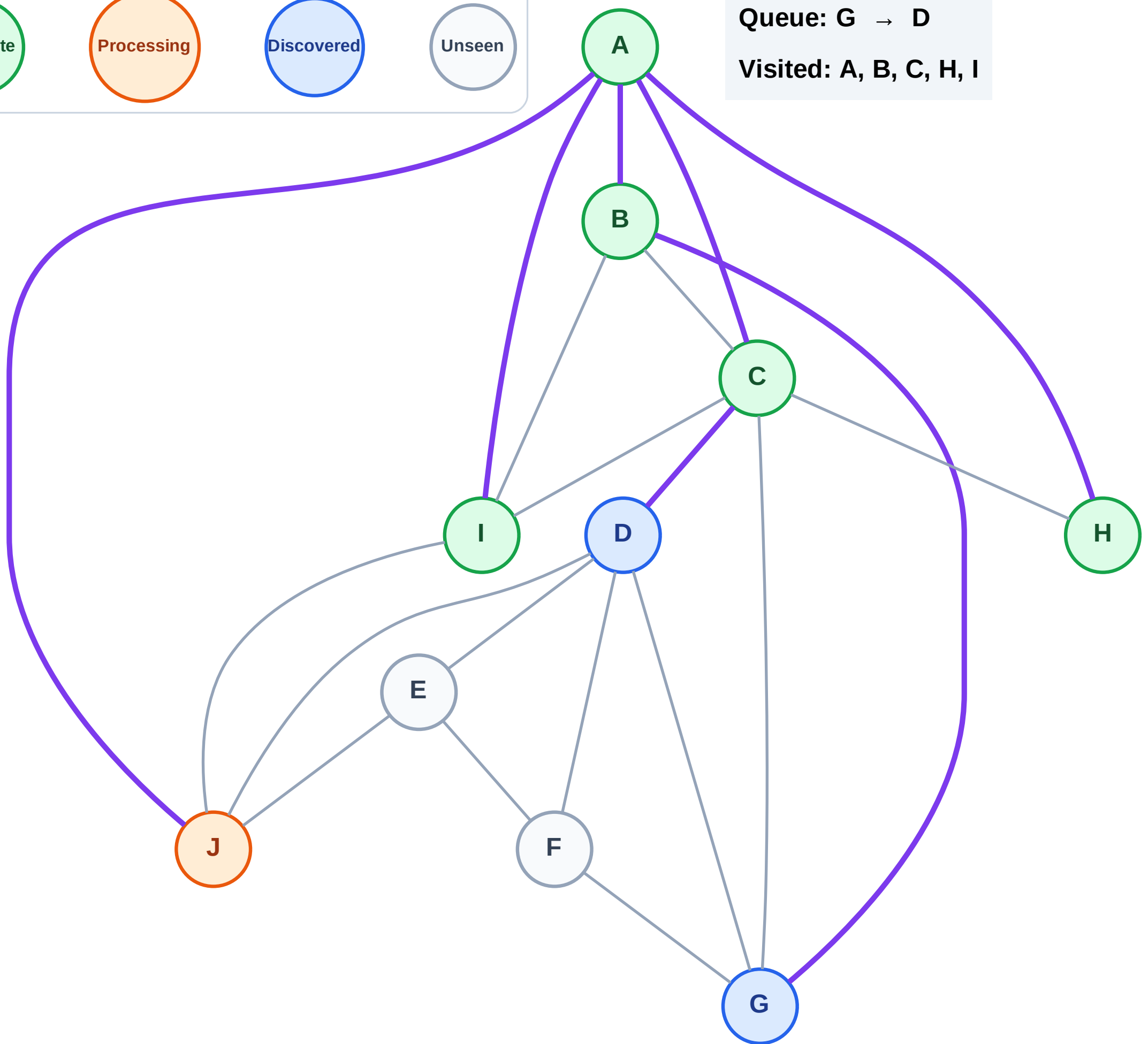
Step 12: Process node J

Legend



Queue: G → D

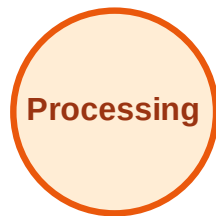
Visited: A, B, C, H, I



BFS Traversal

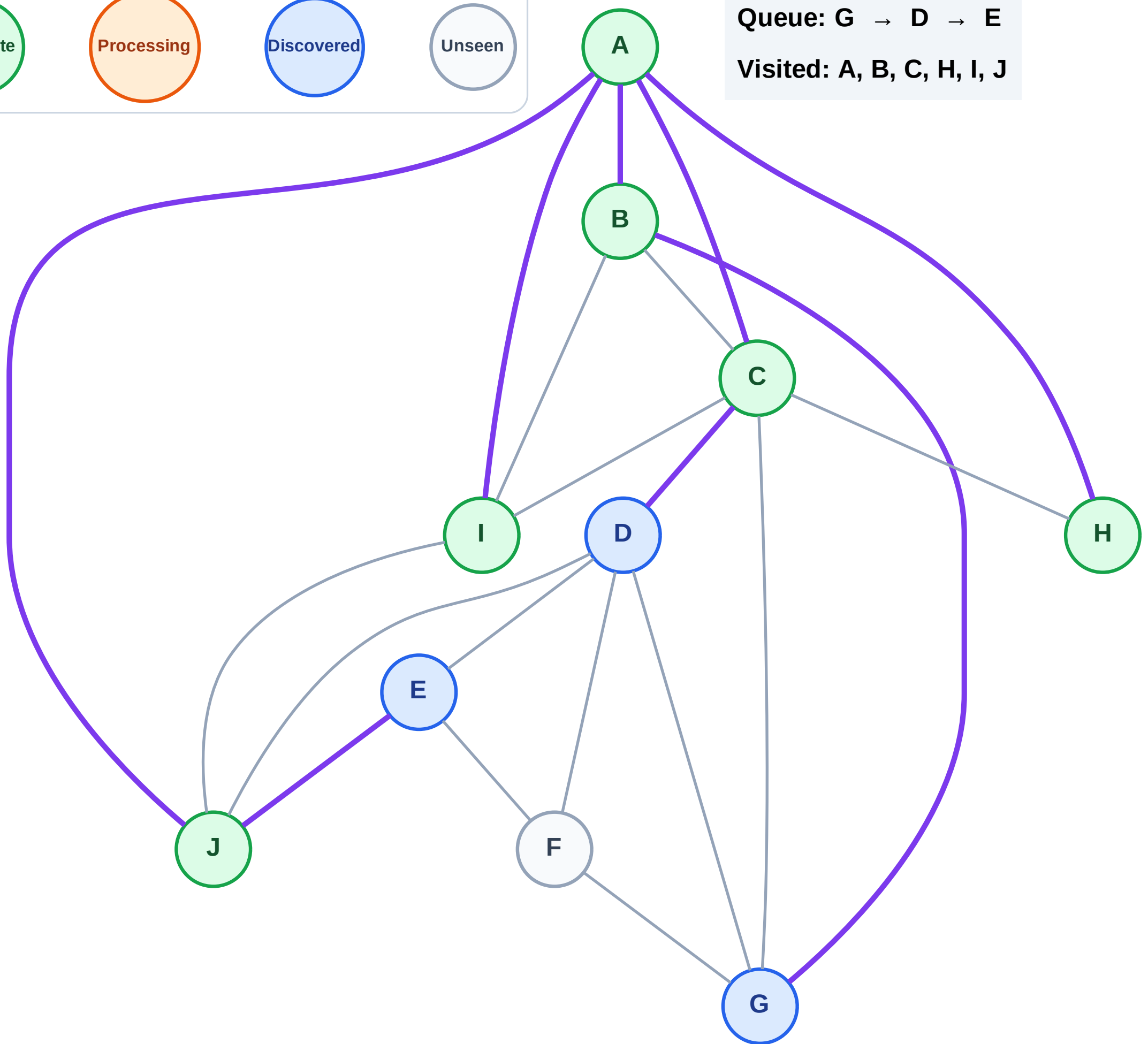
Step 13: Discover E from J

Legend



Queue: G → D → E

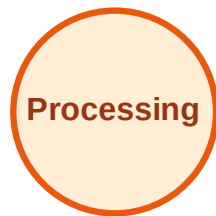
Visited: A, B, C, H, I, J



BFS Traversal

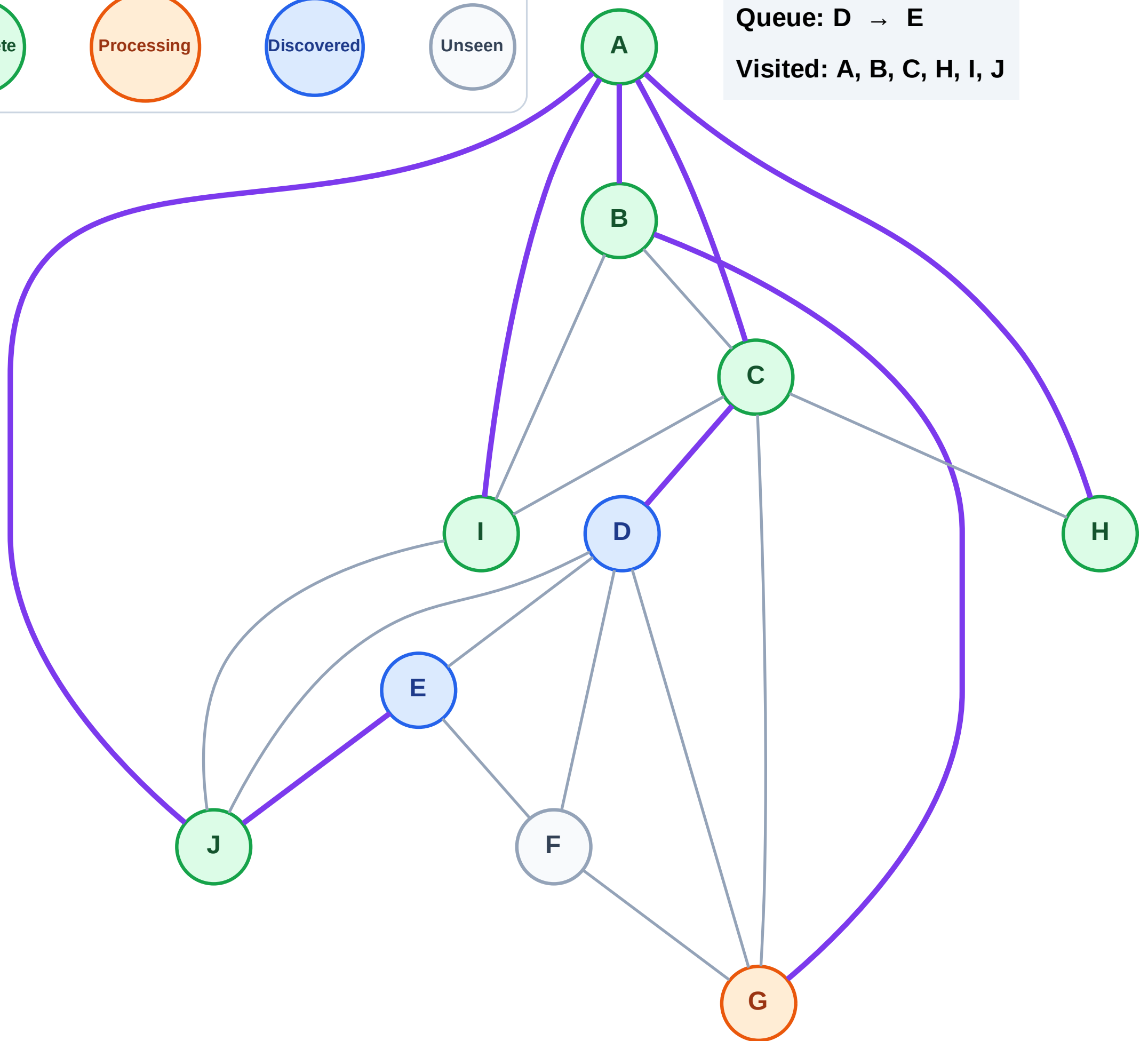
Step 14: Process node G

Legend



Queue: D → E

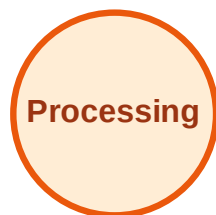
Visited: A, B, C, H, I, J



BFS Traversal

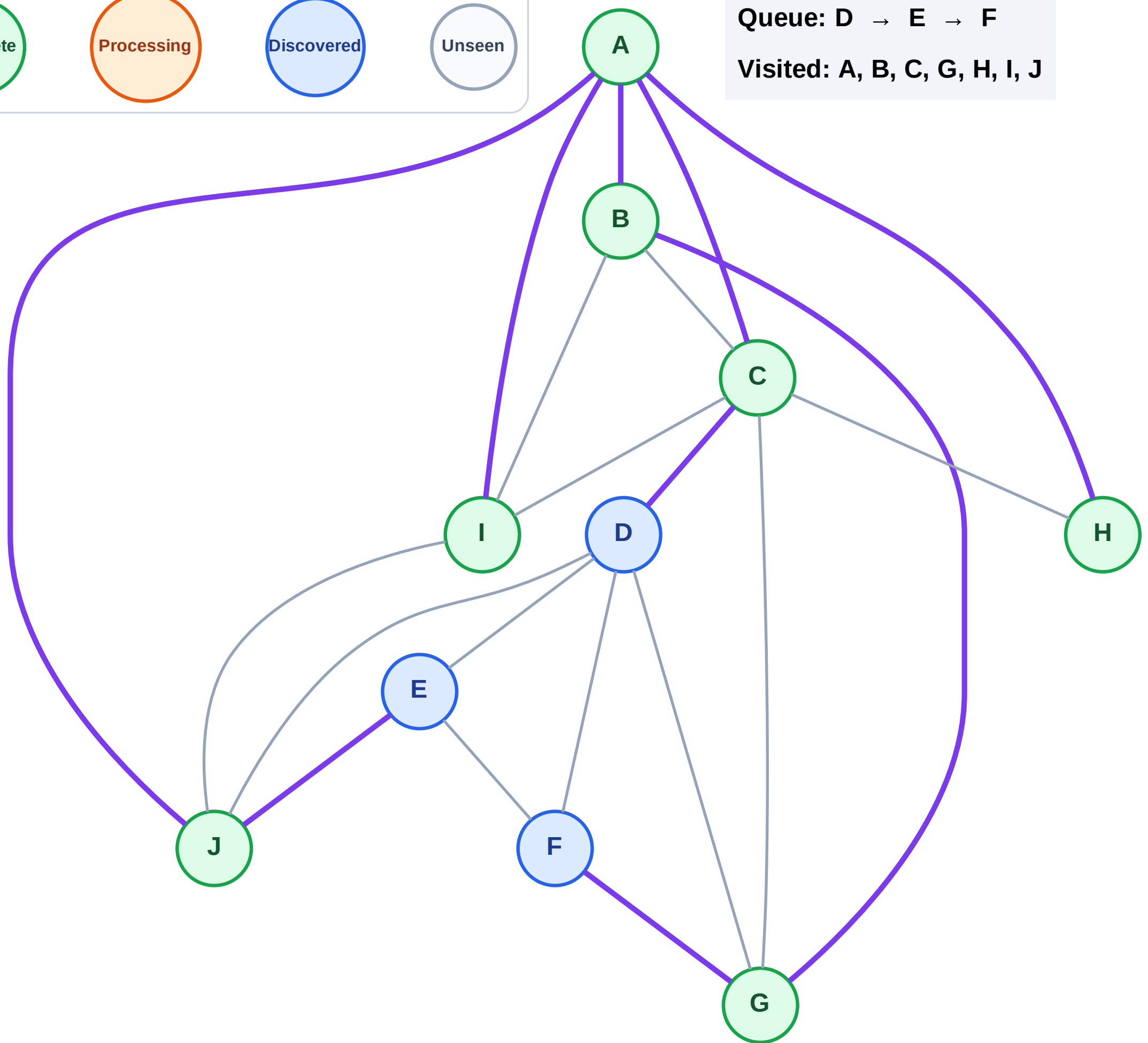
Step 15: Discover F from G

Legend



Queue: D → E → F

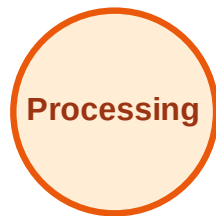
Visited: A, B, C, G, H, I, J



BFS Traversal

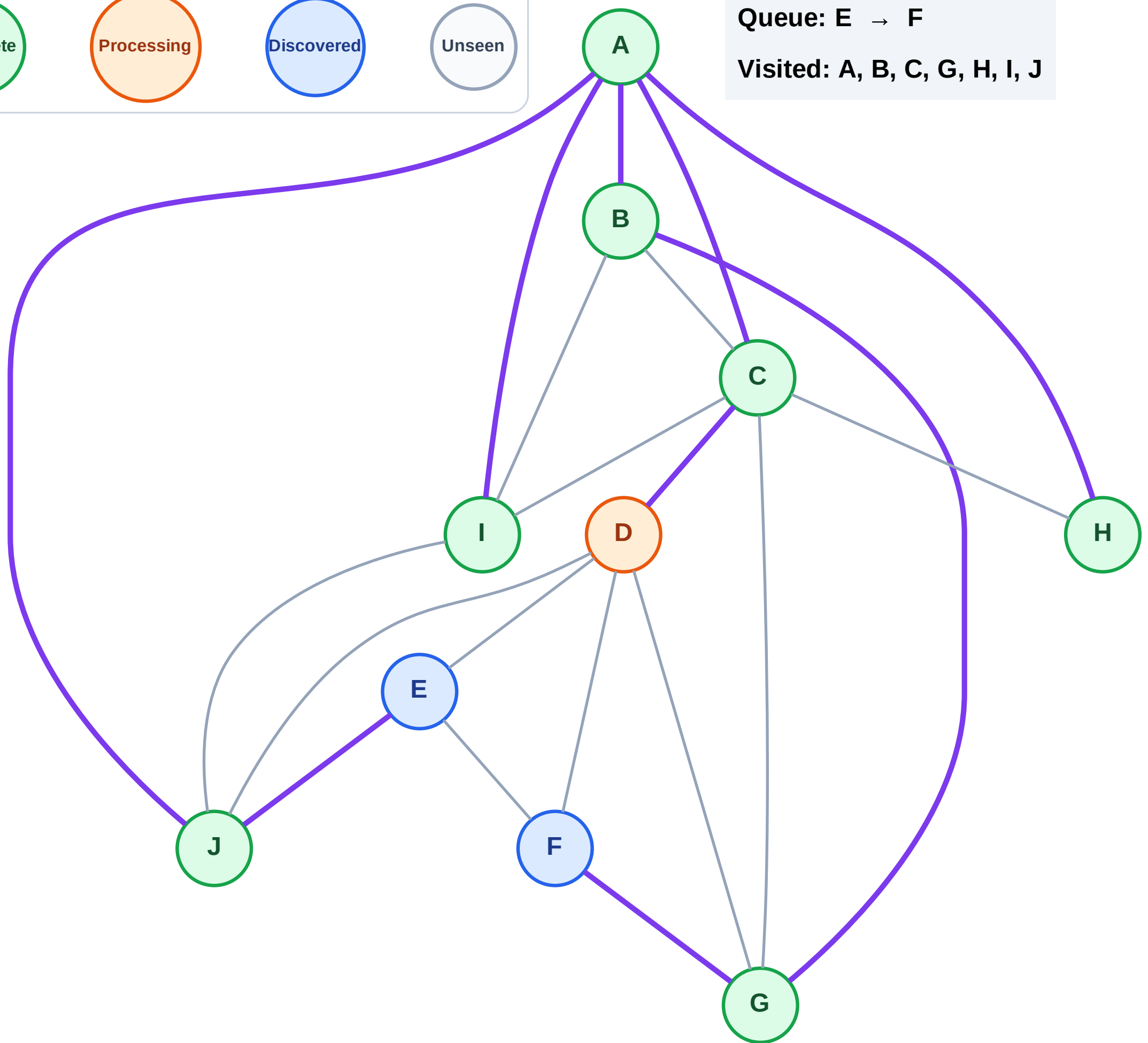
Step 16: Process node D

Legend



Queue: E → F

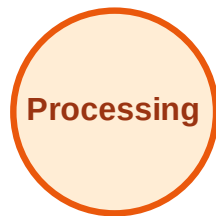
Visited: A, B, C, G, H, I, J



BFS Traversal

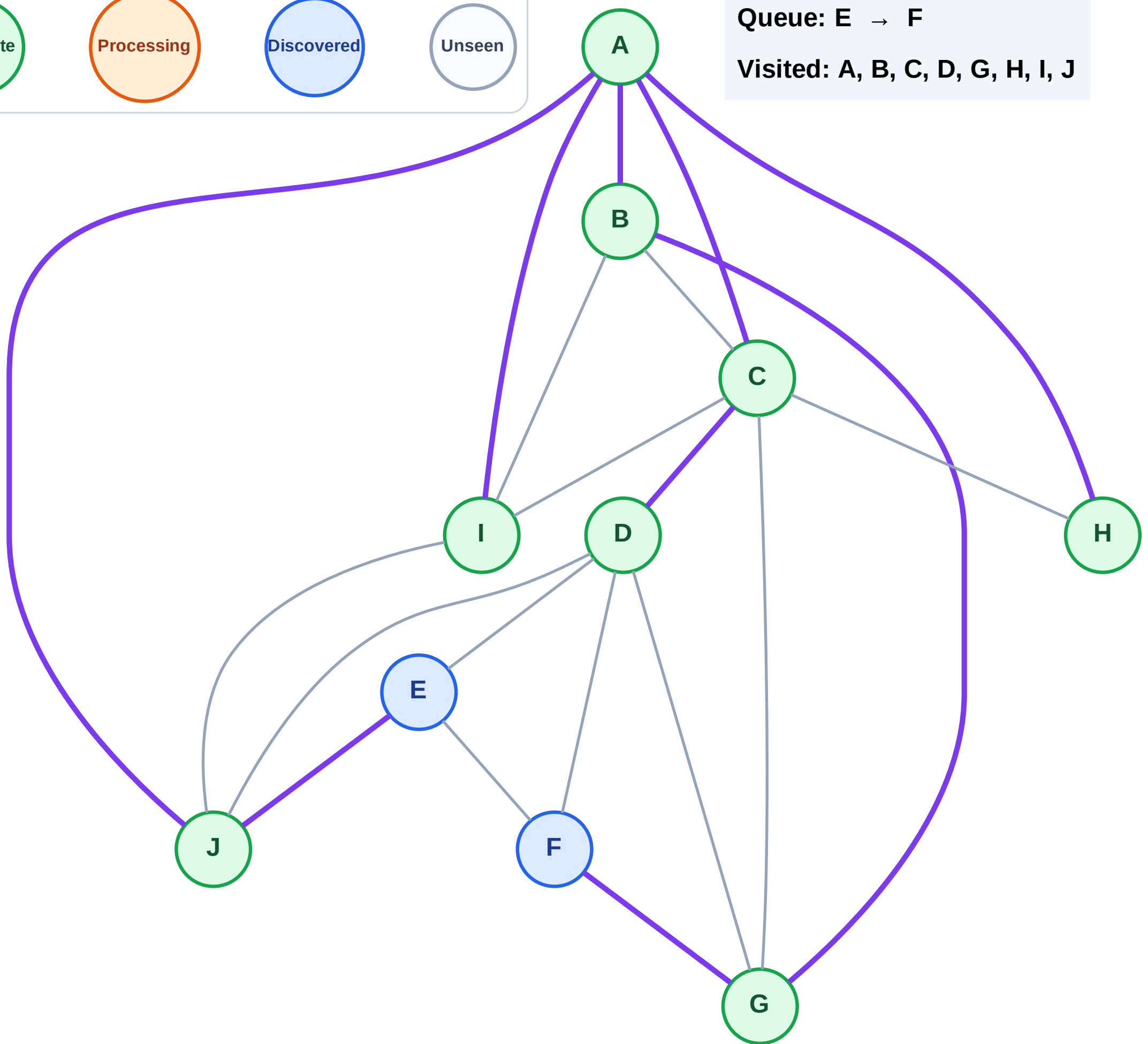
Step 17: No new neighbors from D

Legend



Queue: E → F

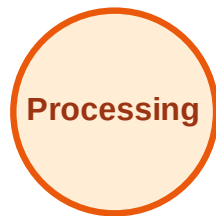
Visited: A, B, C, D, G, H, I, J



BFS Traversal

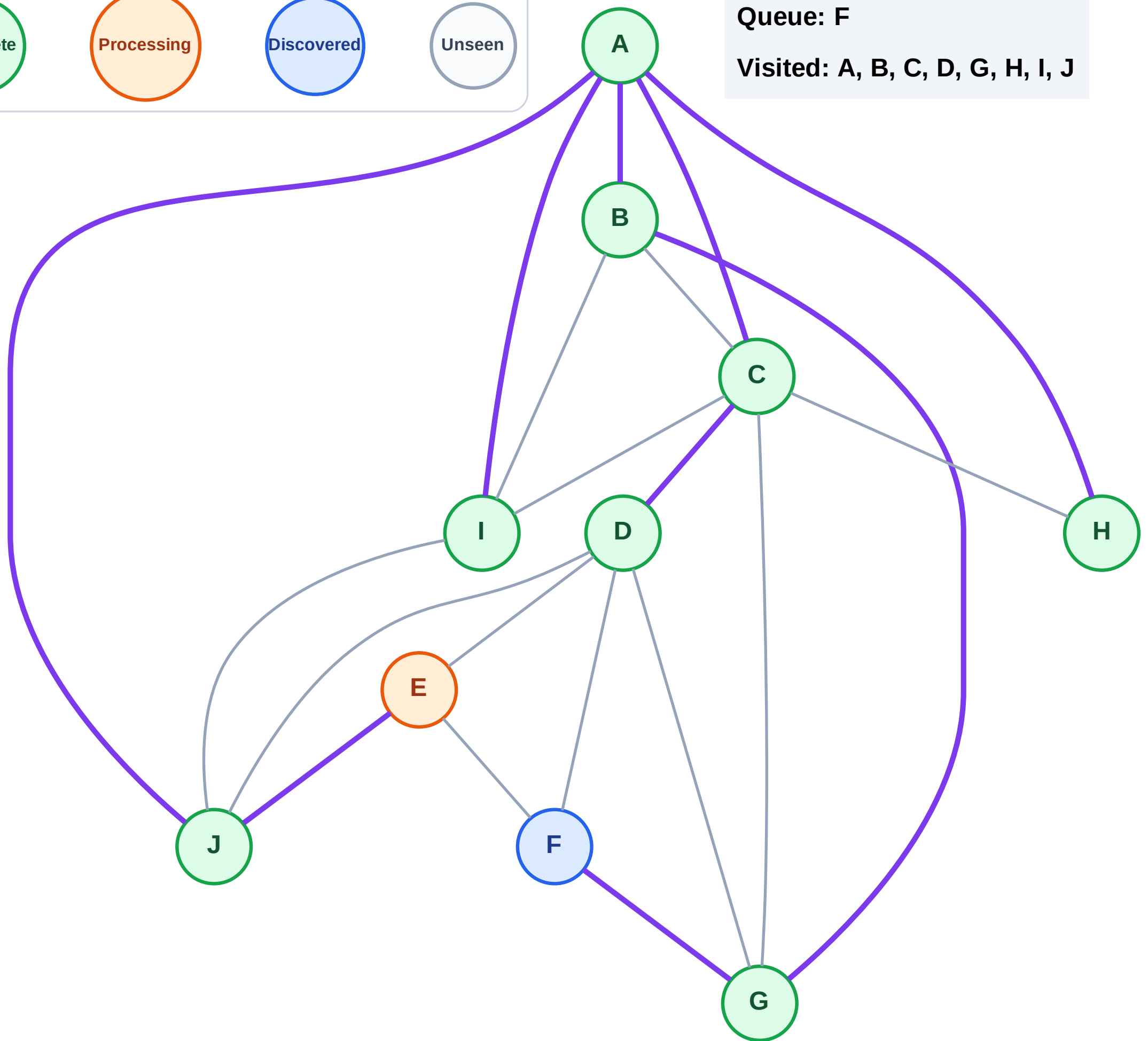
Step 18: Process node E

Legend



Queue: F

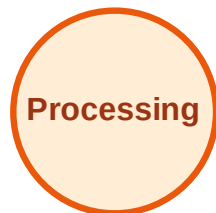
Visited: A, B, C, D, G, H, I, J



BFS Traversal

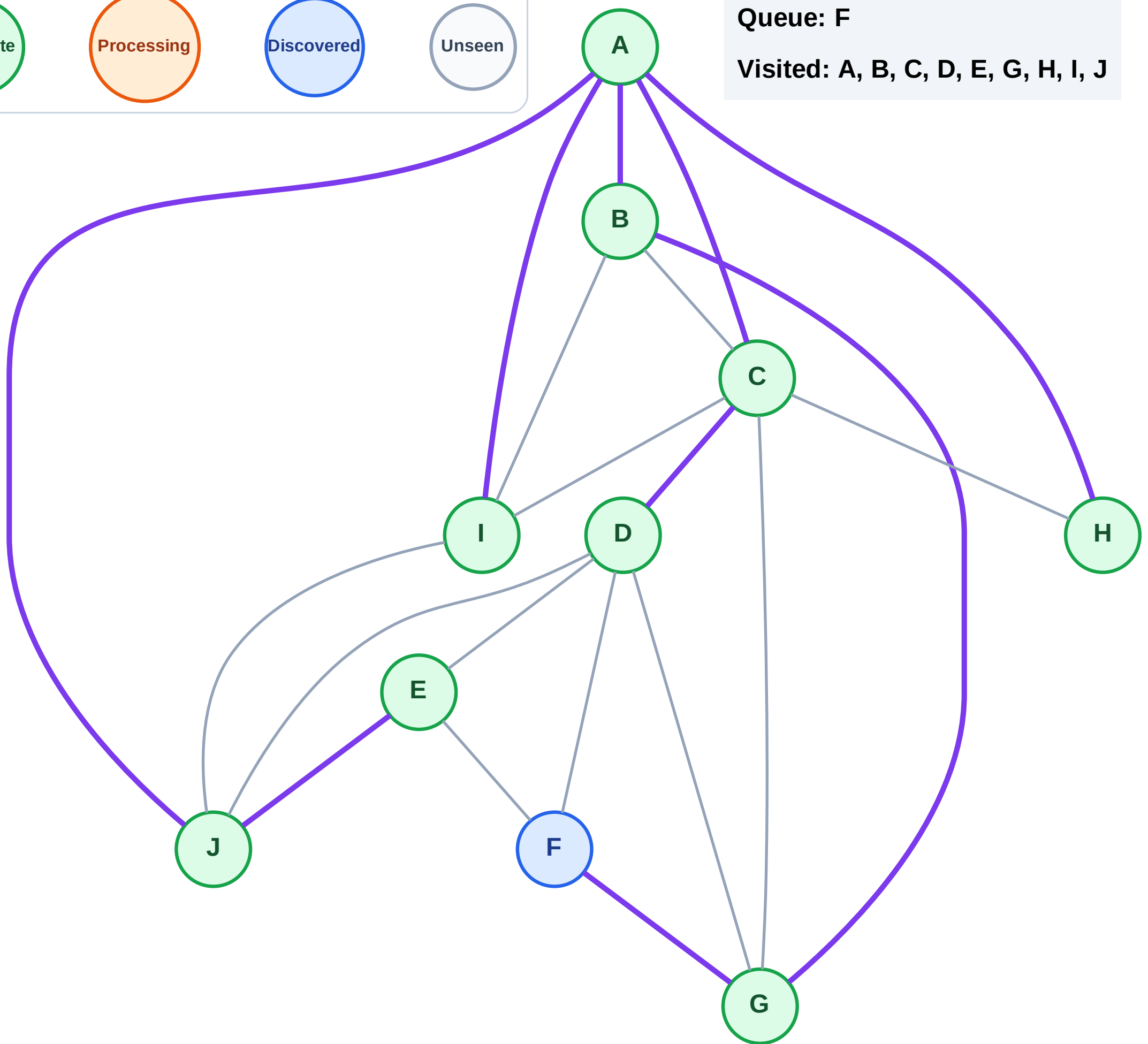
Step 19: No new neighbors from E

Legend



Queue: F

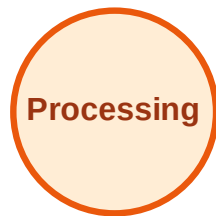
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

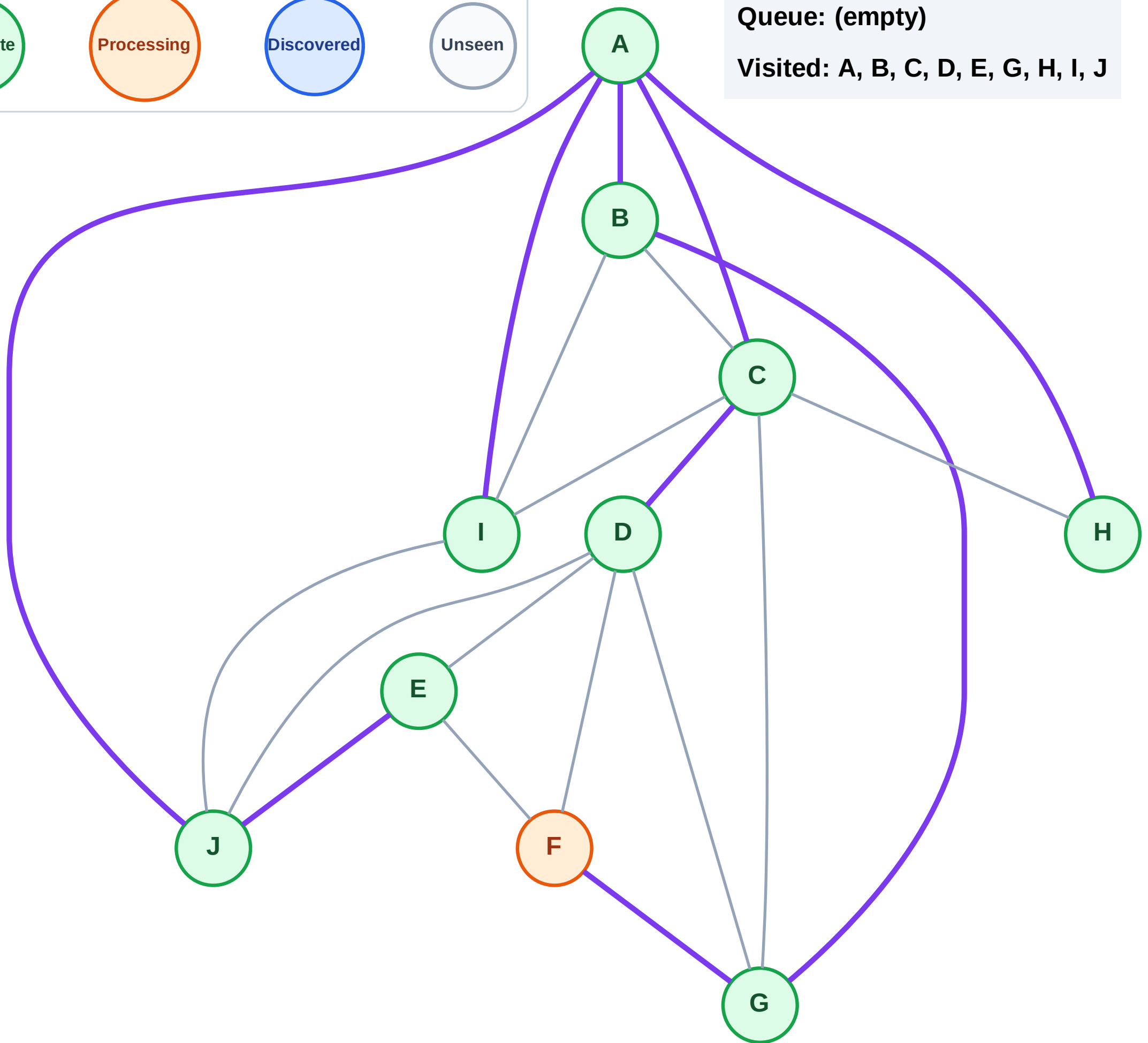
Step 20: Process node F

Legend



Queue: (empty)

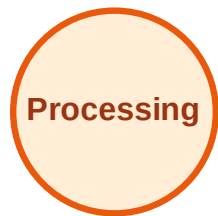
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

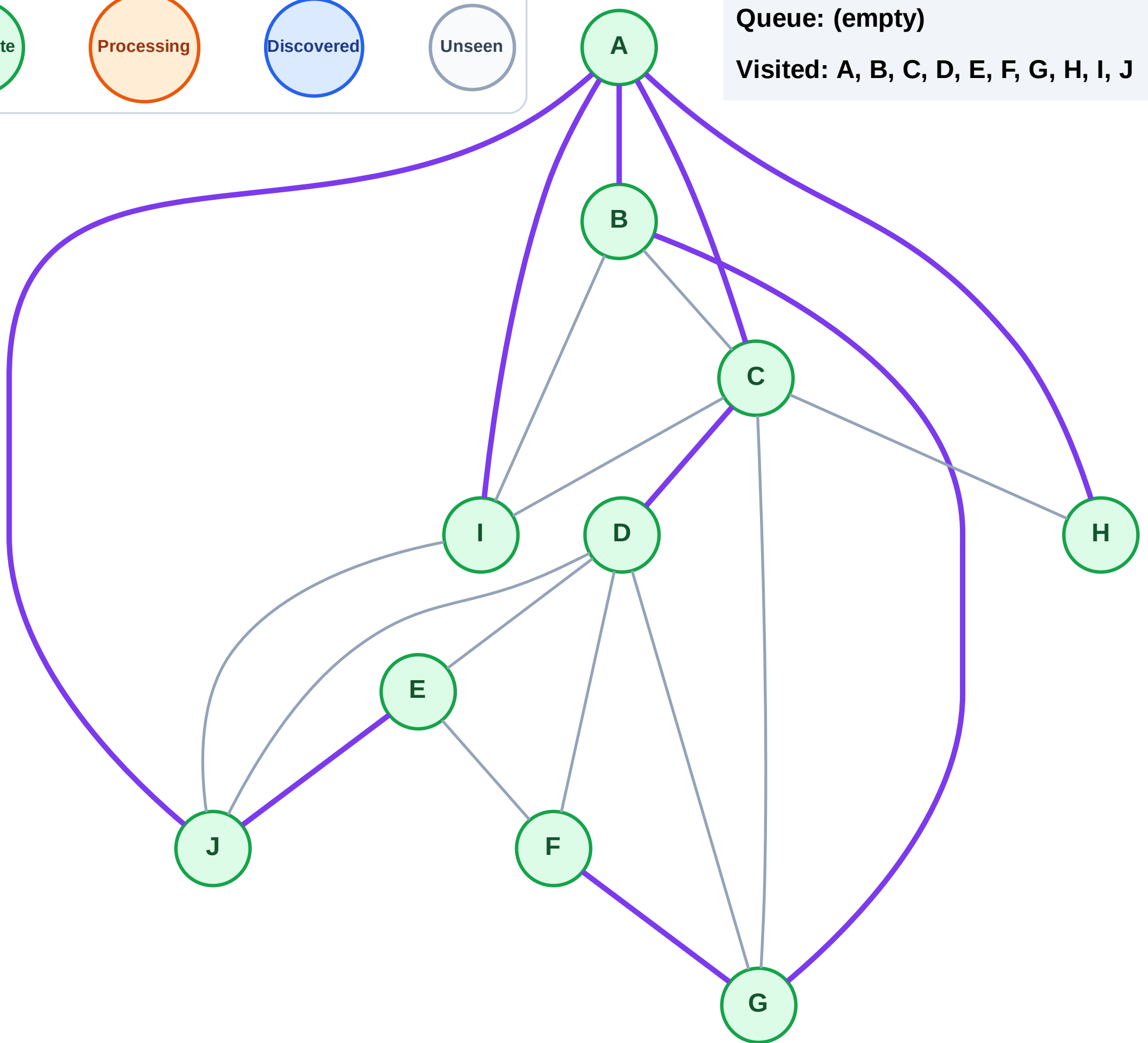
Step 21: No new neighbors from F

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

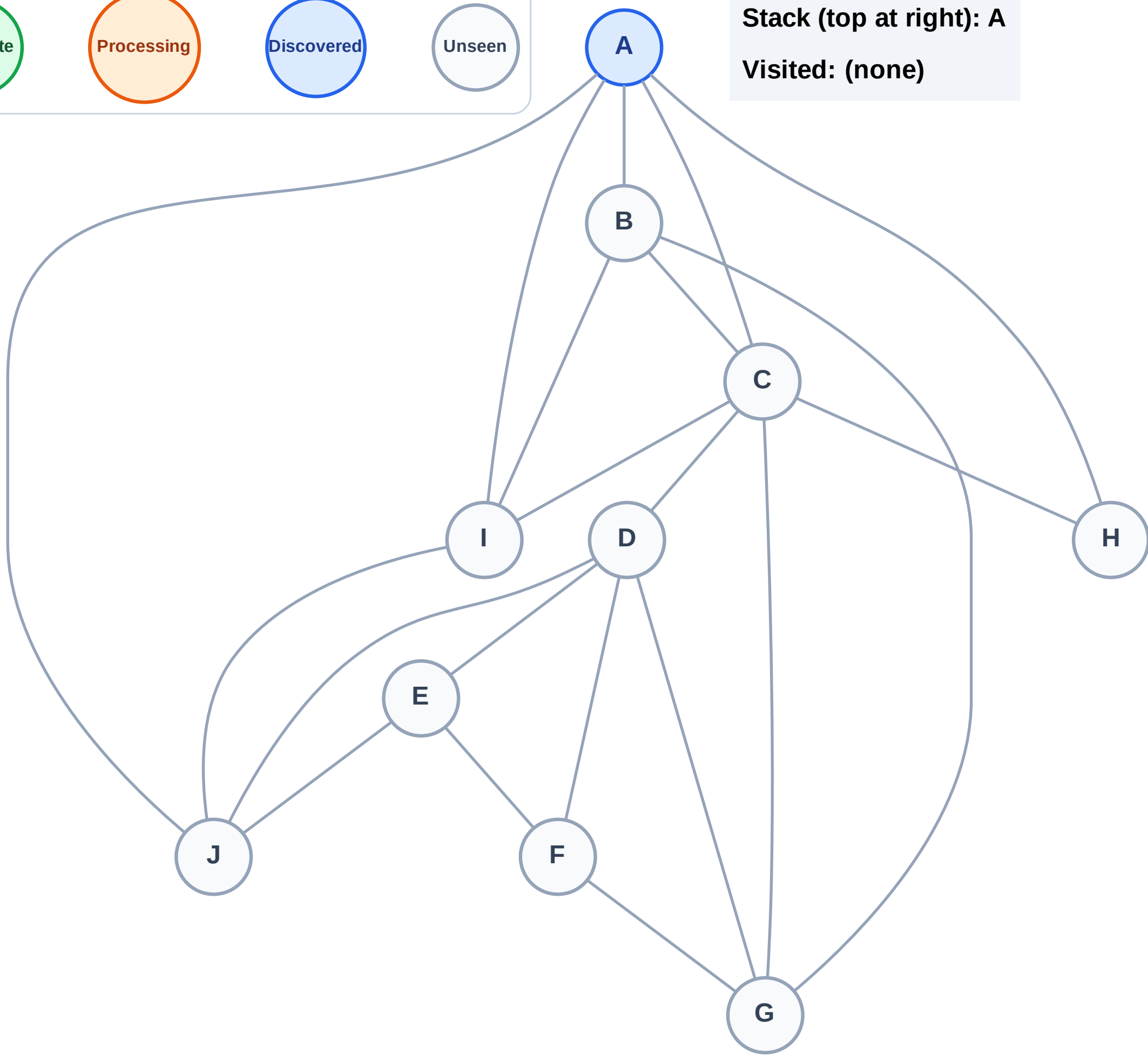
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

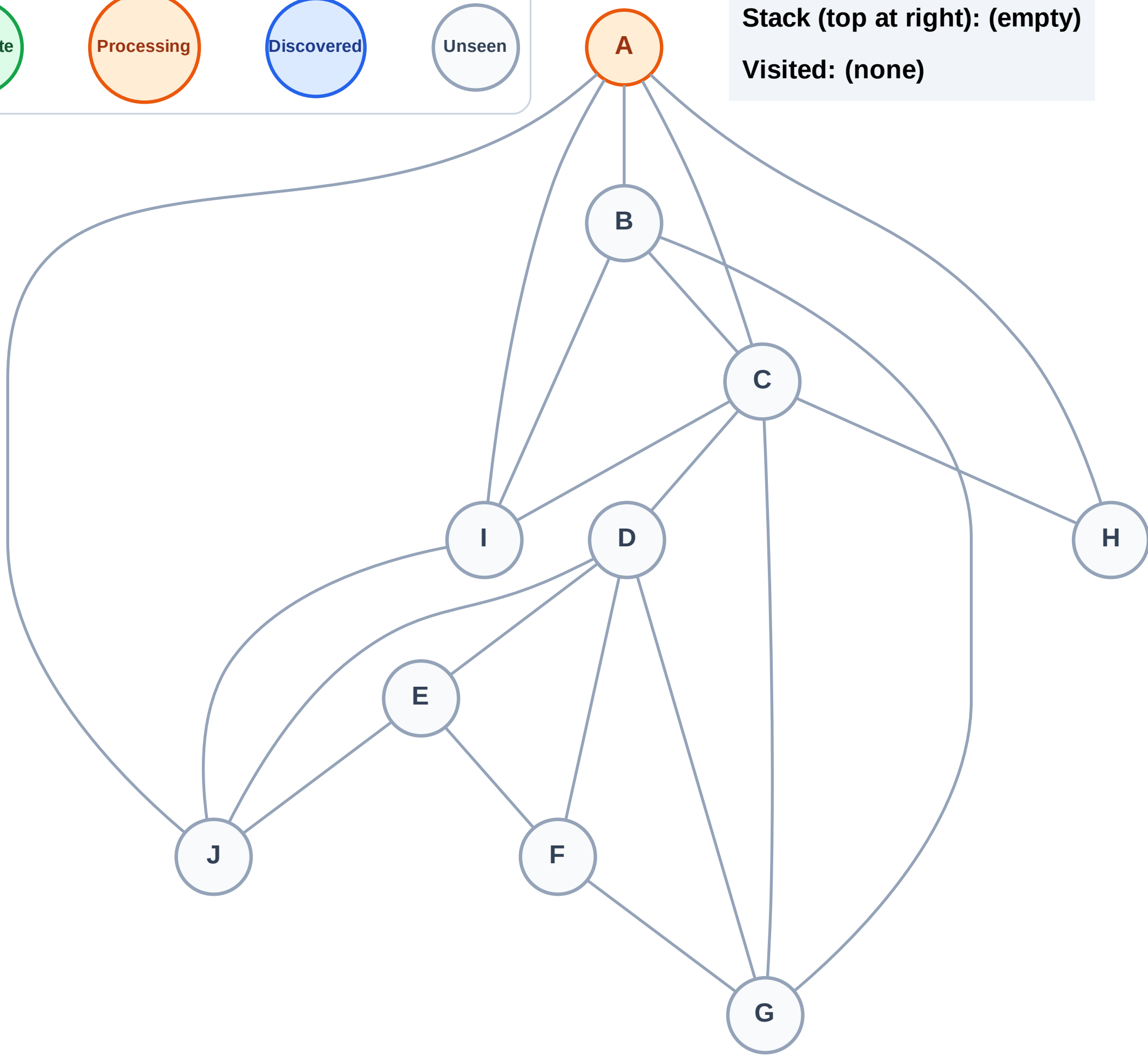
Step 2: Process node A

Legend



Stack (top at right): (empty)

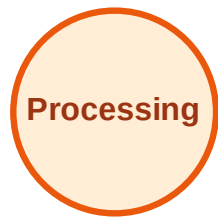
Visited: (none)



DFS Traversal

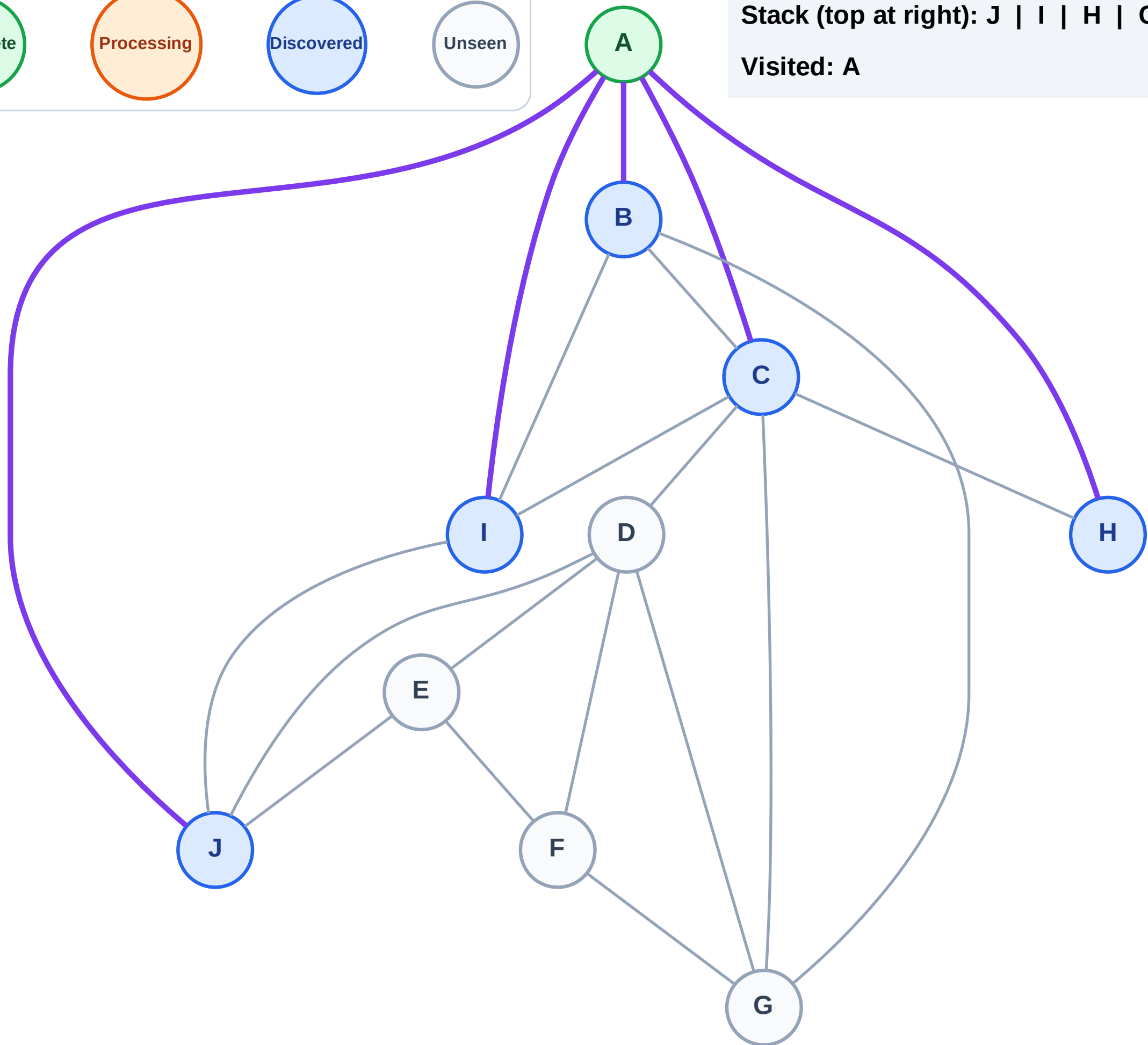
Step 3: Discover J, I, H, C, B from A

Legend



Stack (top at right): J | I | H | C | B

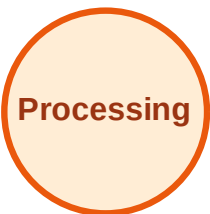
Visited: A



DFS Traversal

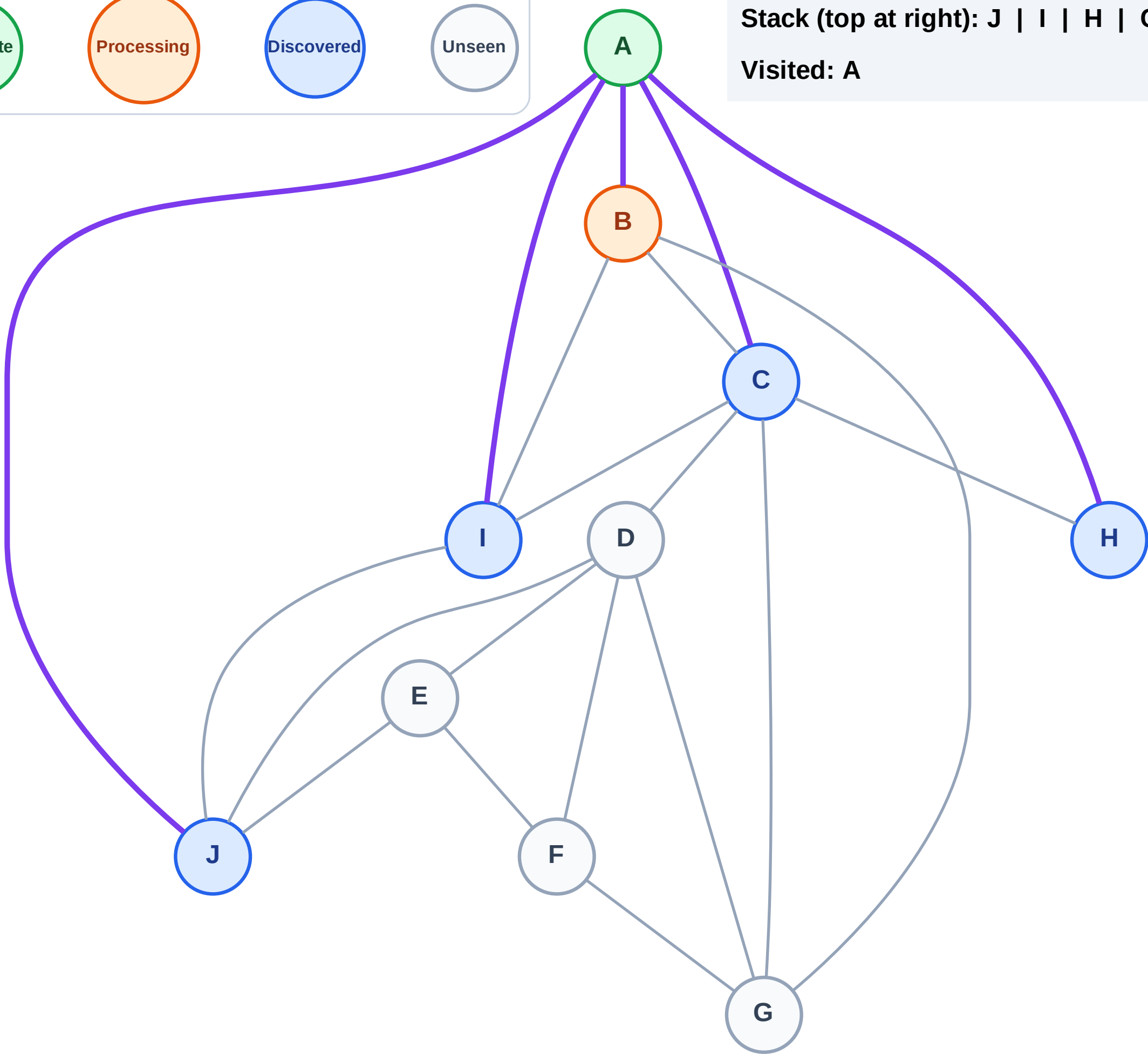
Step 4: Process node B

Legend



Stack (top at right): J | I | H | C

Visited: A



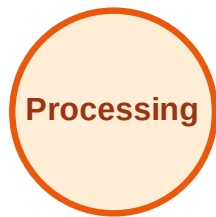
DFS Traversal

Step 5: Discover G from B

Legend



Complete



Processing



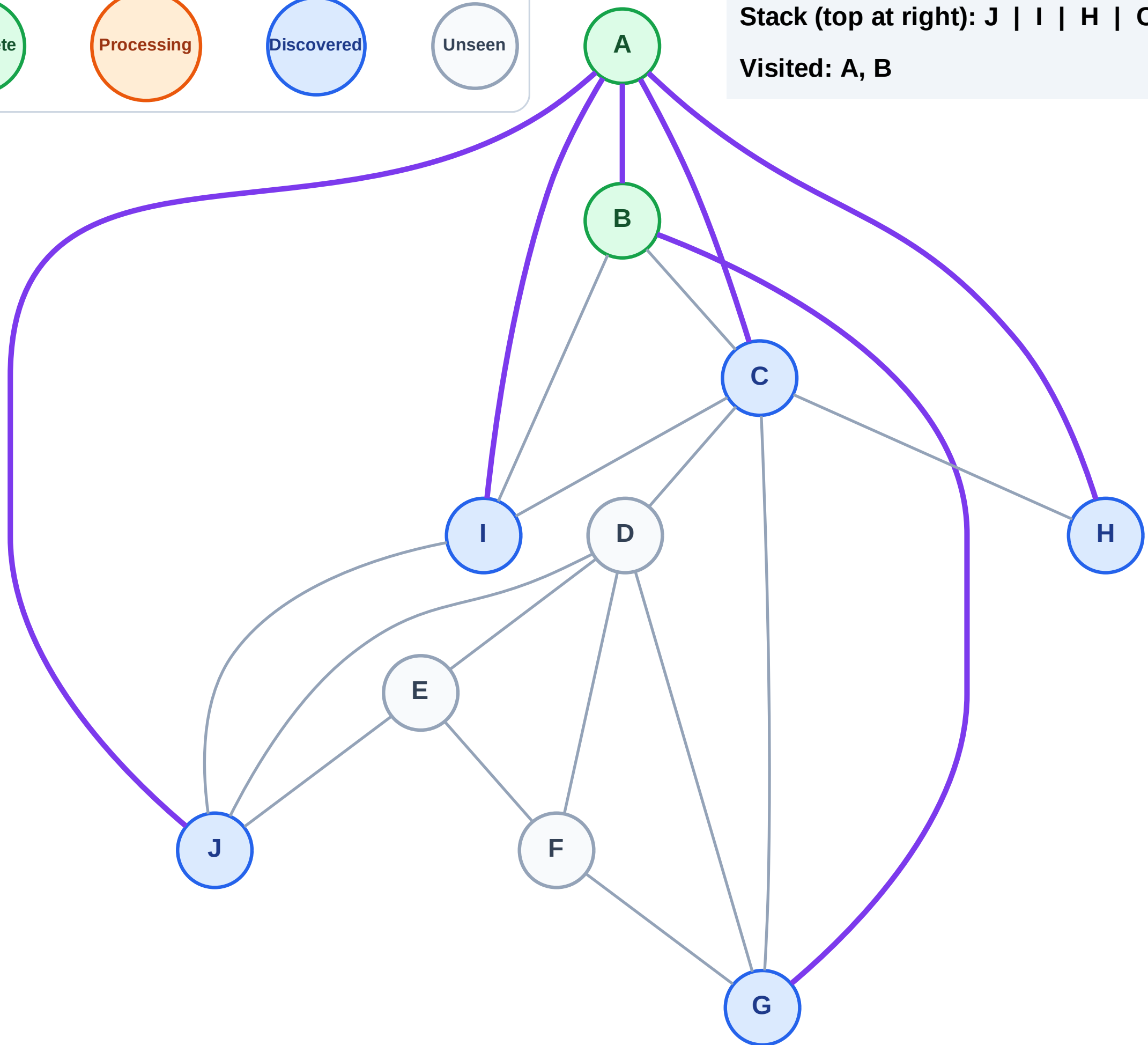
Discovered



Unseen

Stack (top at right): J | I | H | C | G

Visited: A, B



DFS Traversal

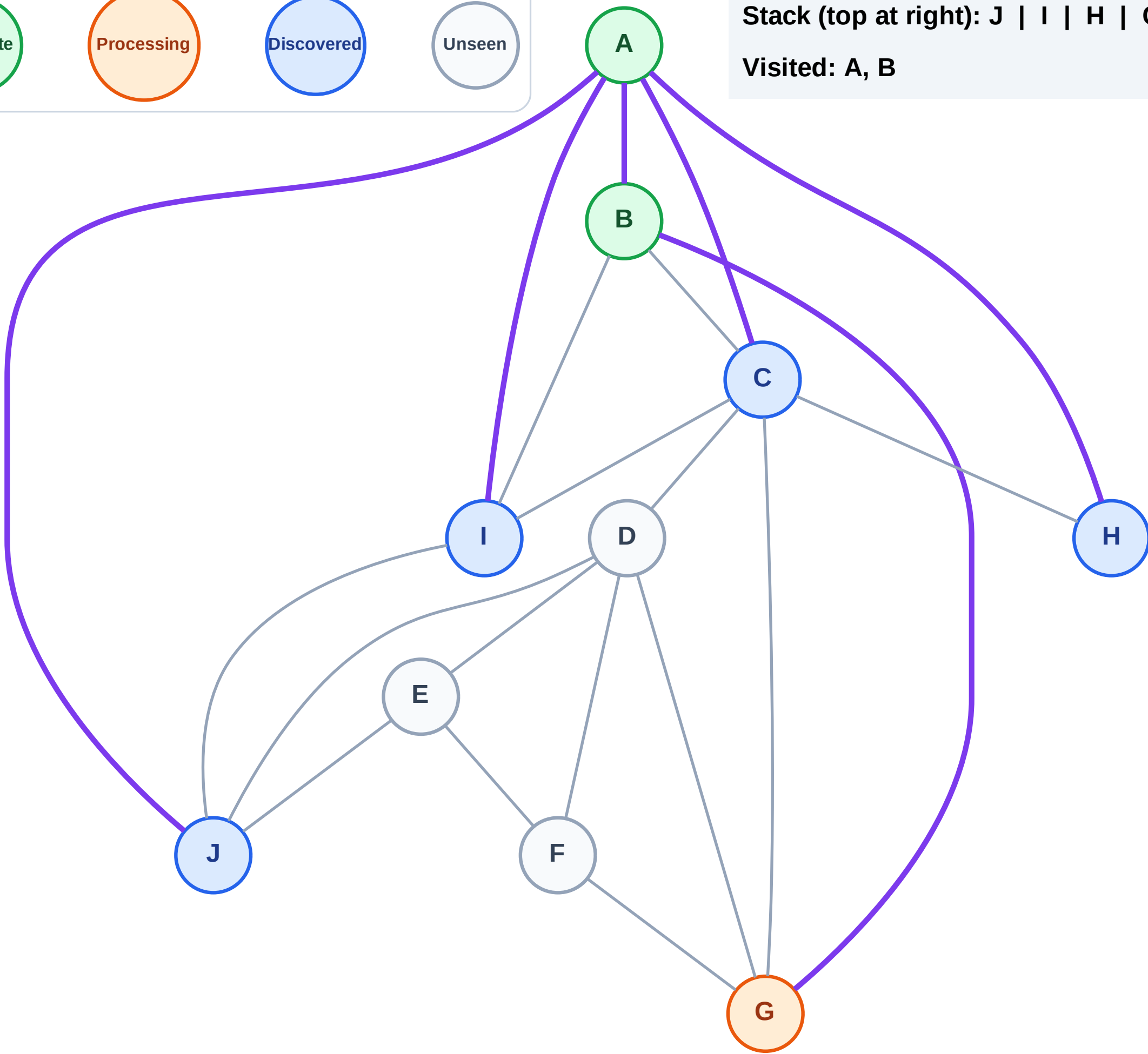
Step 6: Process node G

Legend



Stack (top at right): J | I | H | C

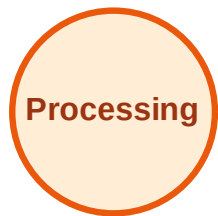
Visited: A, B



DFS Traversal

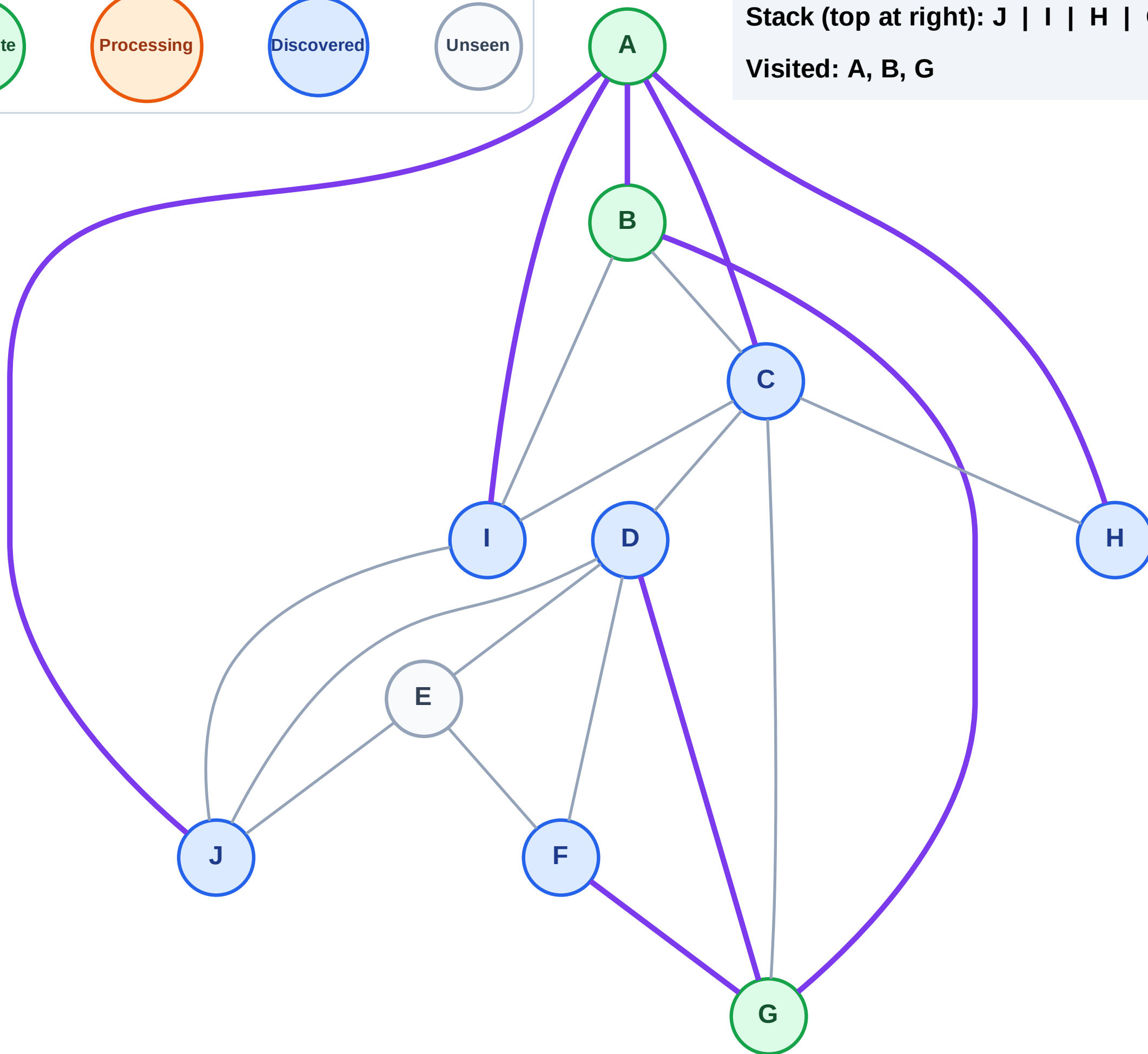
Step 7: Discover F, D from G

Legend



Stack (top at right): J | I | H | C | F | D

Visited: A, B, G



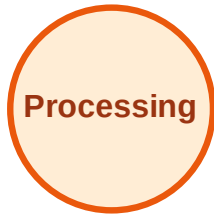
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



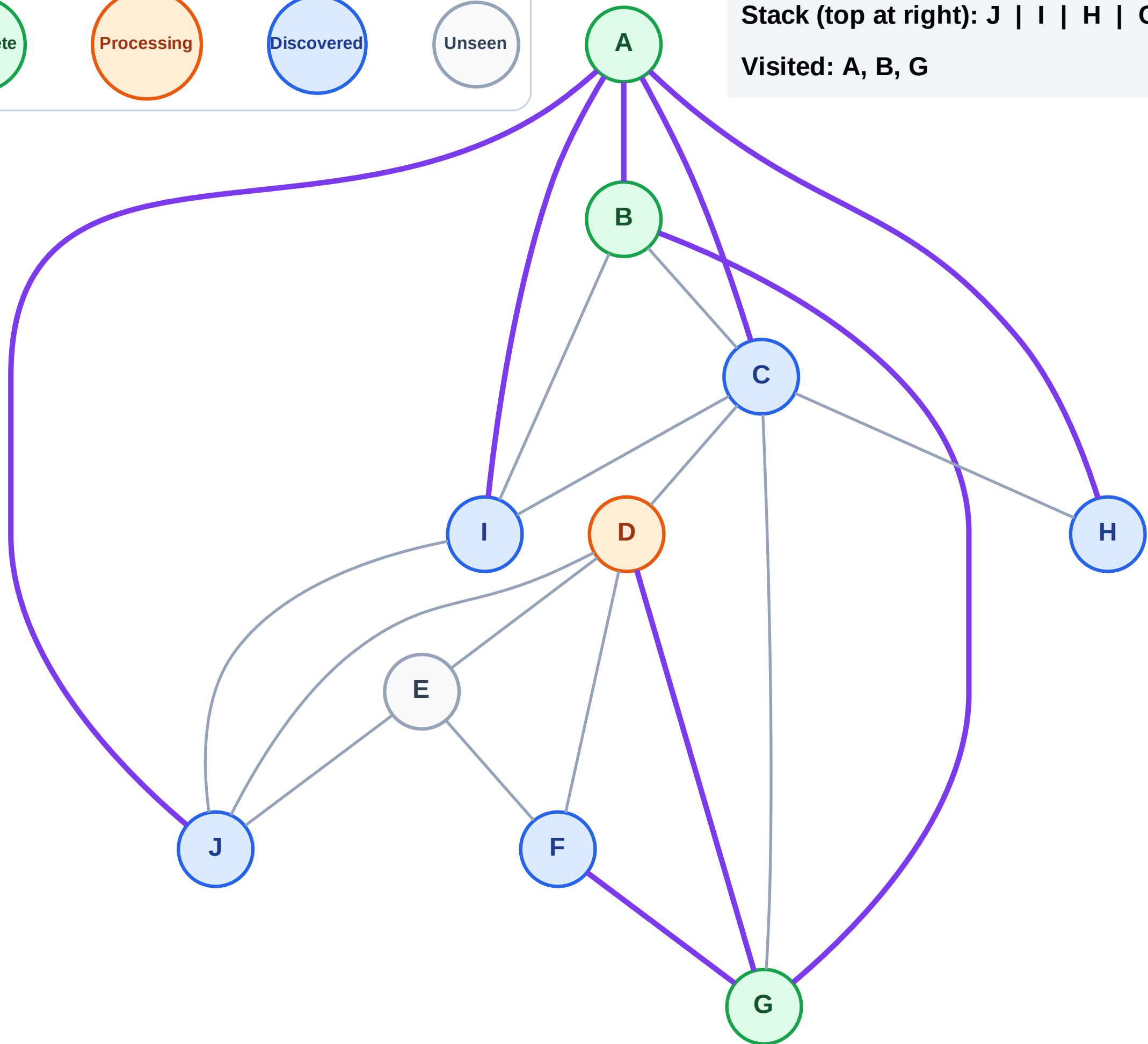
Discovered



Unseen

Stack (top at right): J | I | H | C | F

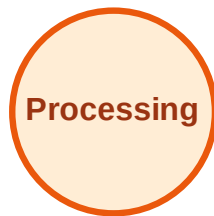
Visited: A, B, G



DFS Traversal

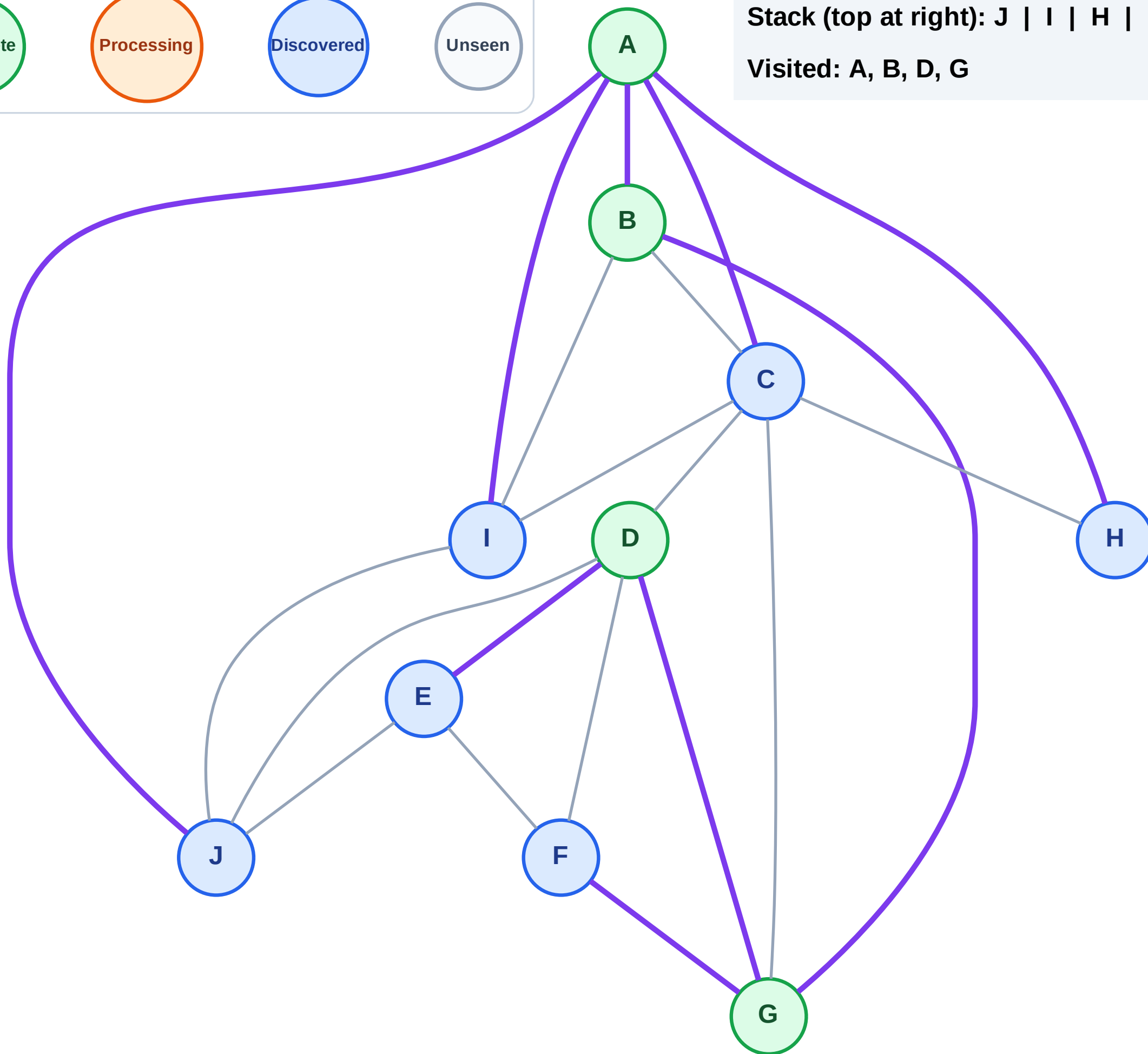
Step 9: Discover E from D

Legend



Stack (top at right): J | I | H | C | F | E

Visited: A, B, D, G



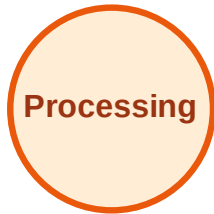
DFS Traversal

Step 10: Process node E

Legend



Complete



Processing



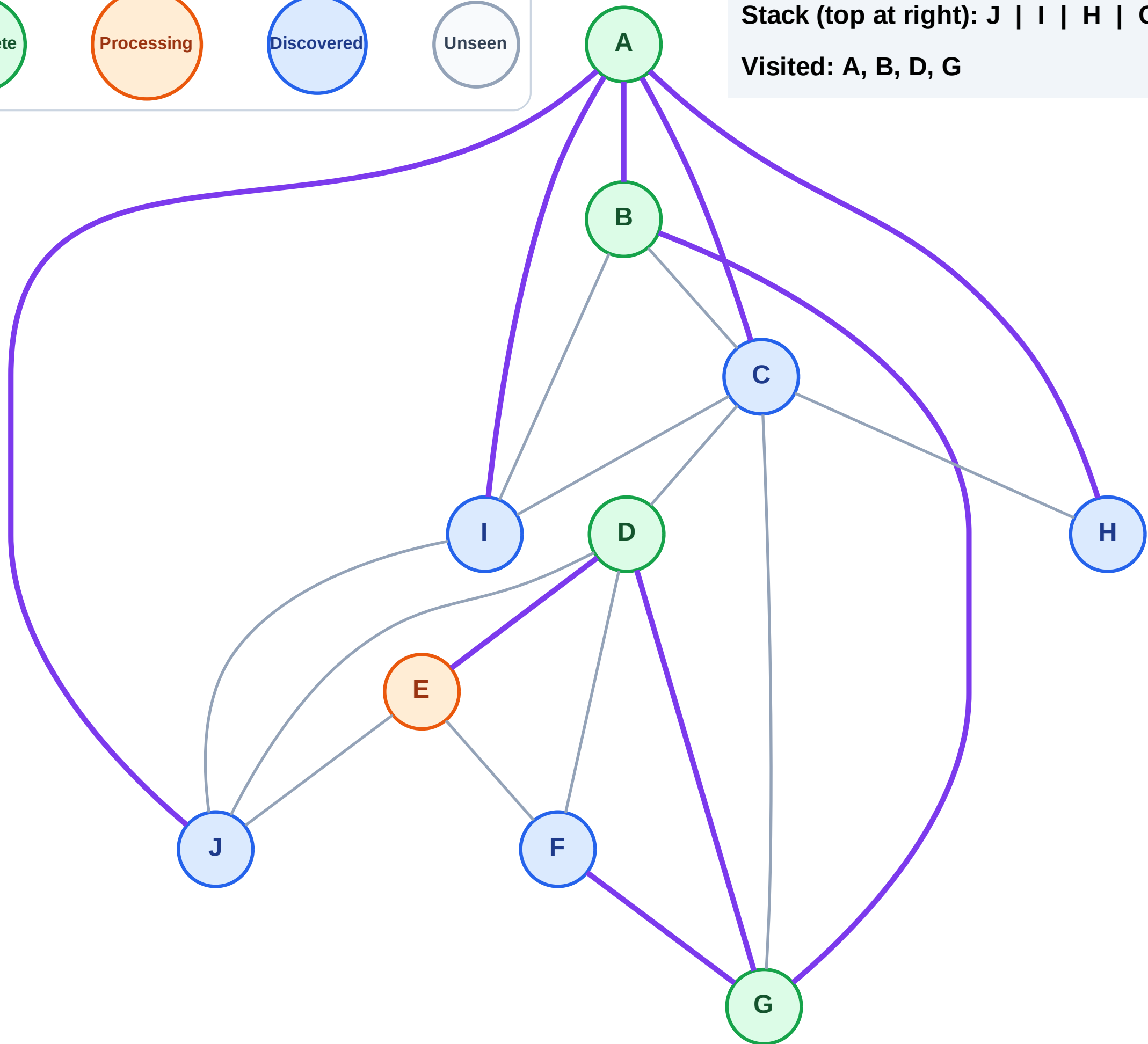
Discovered



Unseen

Stack (top at right): J | I | H | C | F

Visited: A, B, D, G



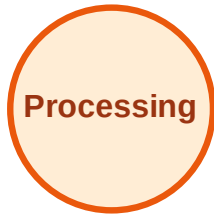
DFS Traversal

Step 11: No new neighbors from E

Legend



Complete



Processing



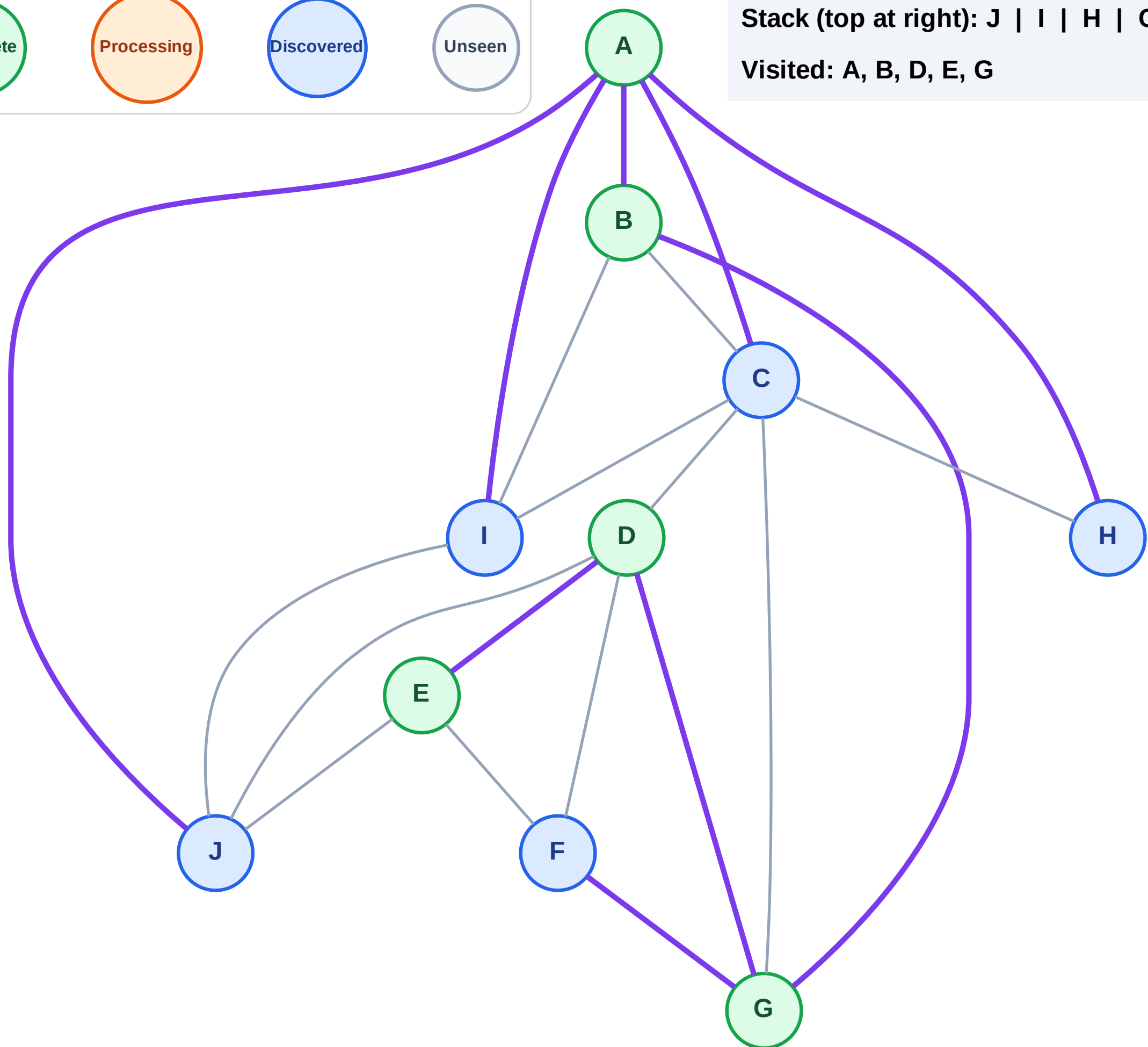
Discovered



Unseen

Stack (top at right): J | I | H | C | F

Visited: A, B, D, E, G



DFS Traversal

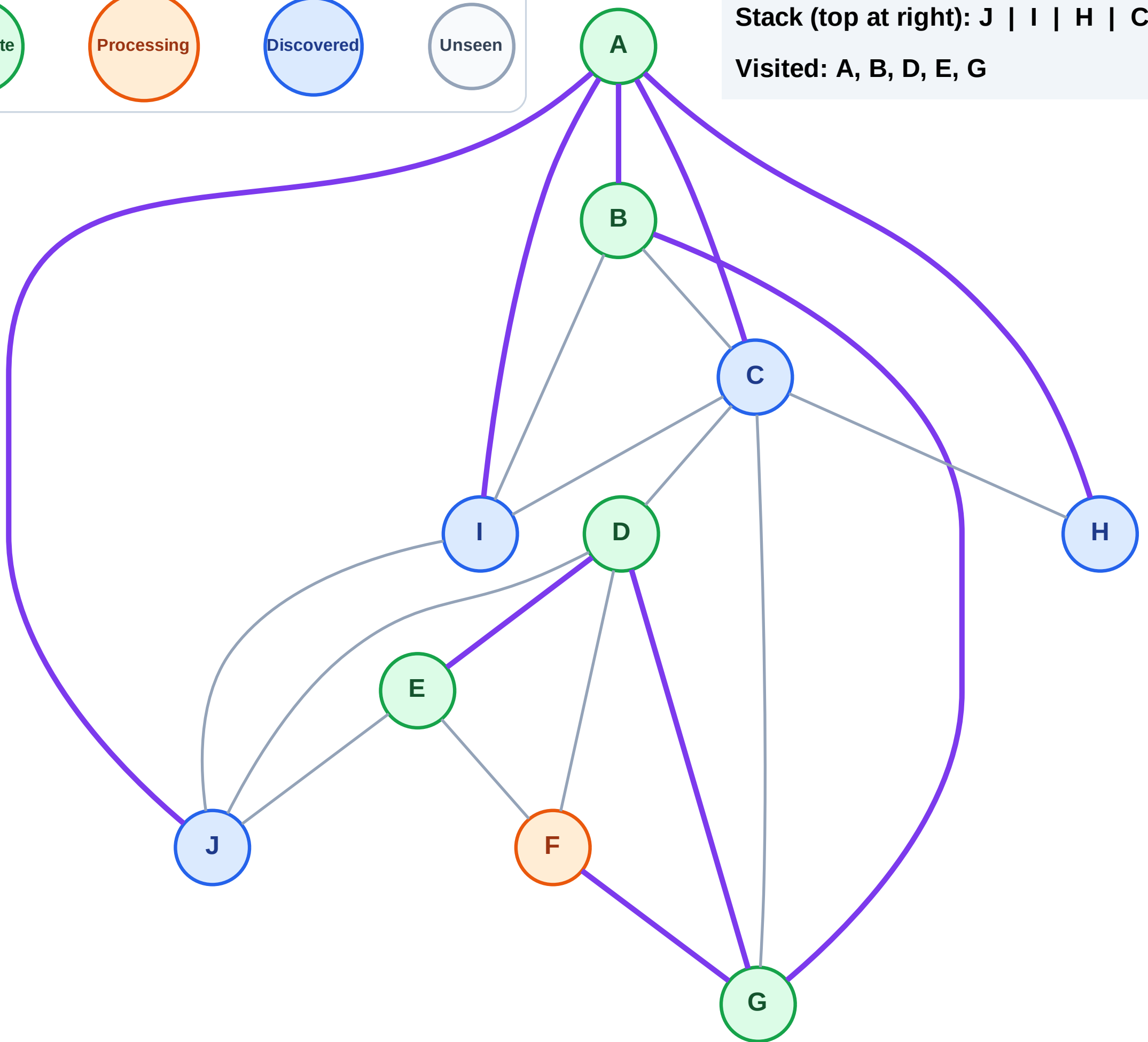
Step 12: Process node F

Legend



Stack (top at right): J | I | H | C

Visited: A, B, D, E, G



DFS Traversal

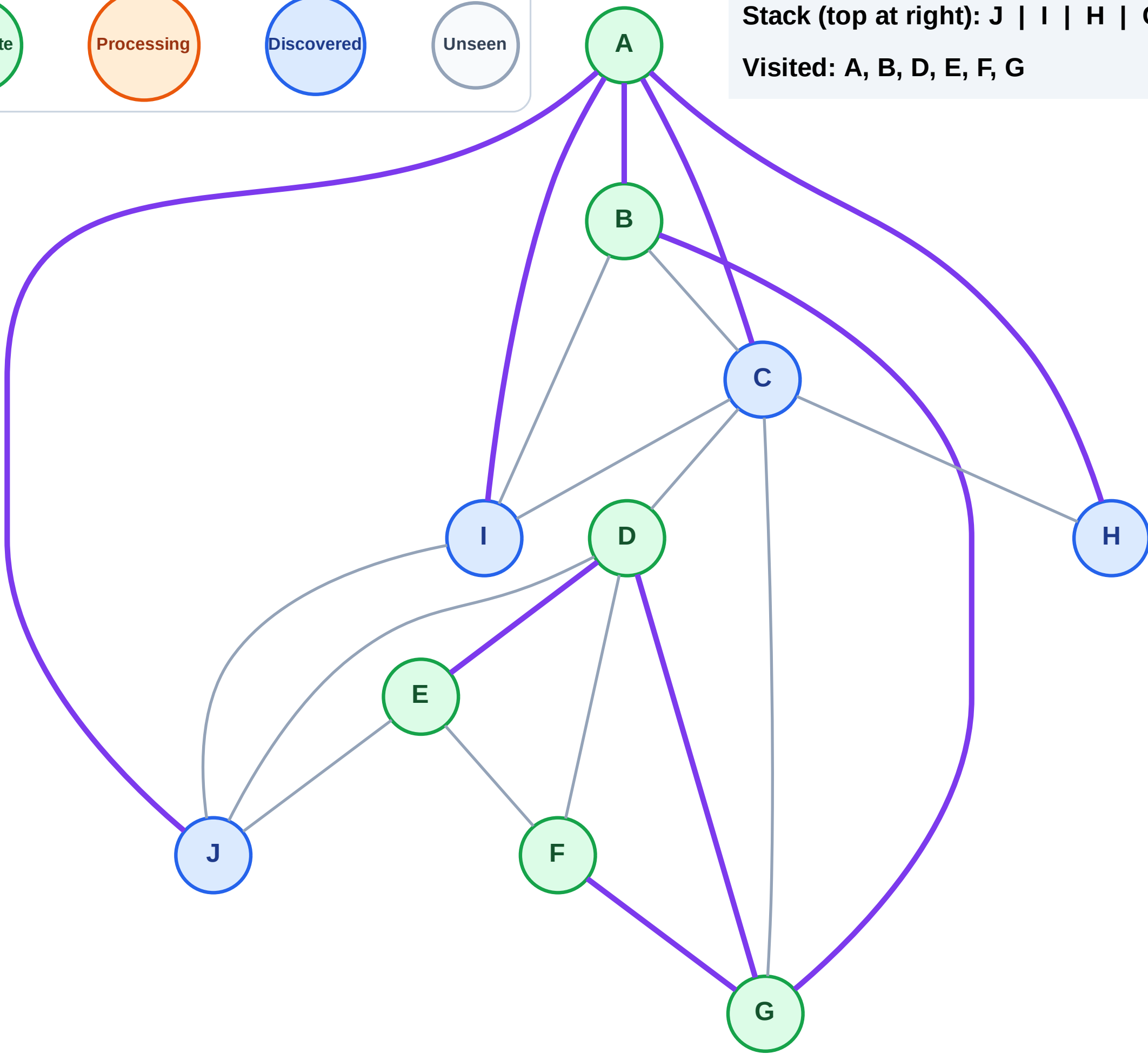
Step 13: No new neighbors from F

Legend



Stack (top at right): J | I | H | C

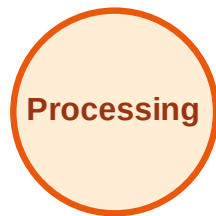
Visited: A, B, D, E, F, G



DFS Traversal

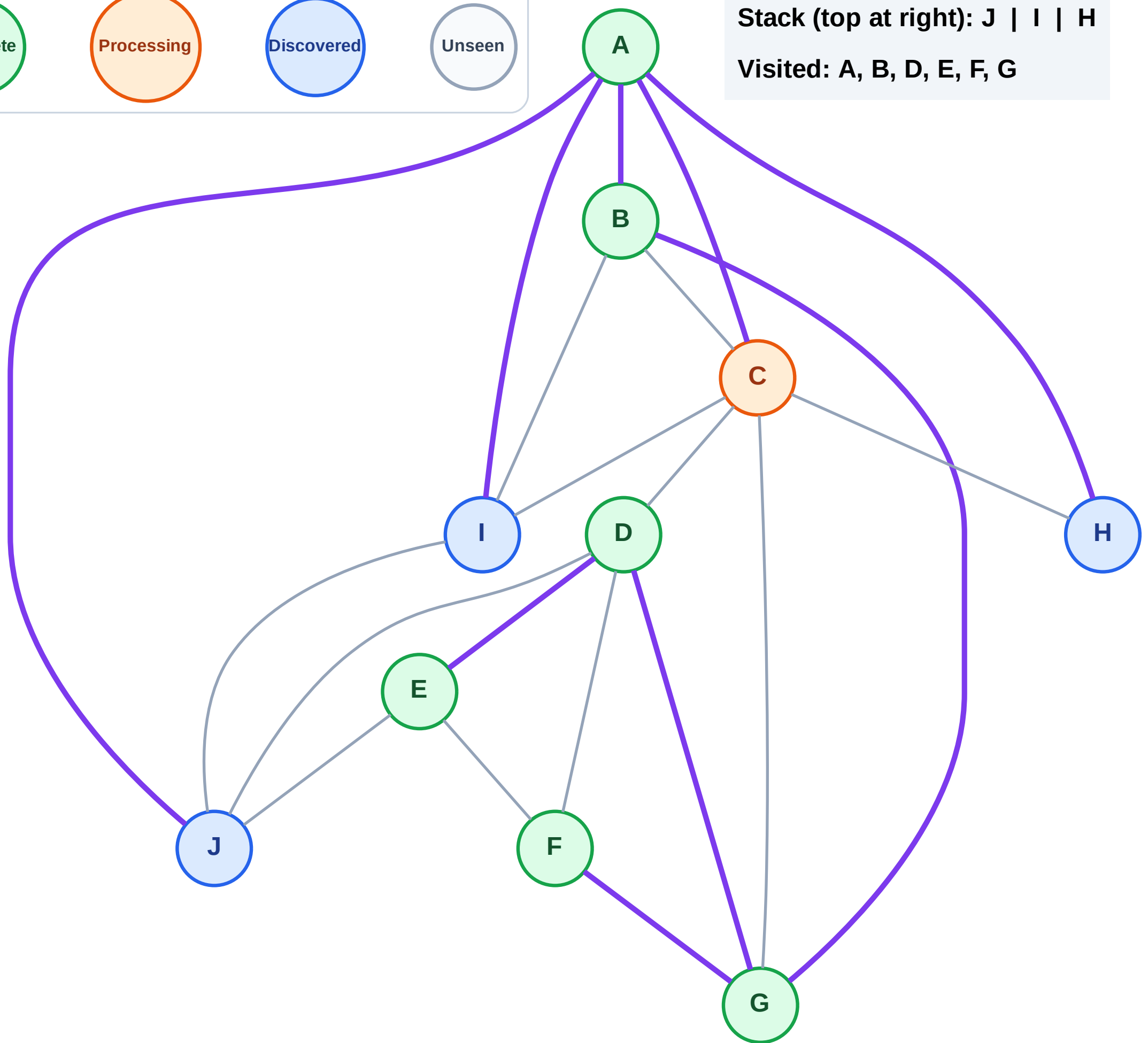
Step 14: Process node C

Legend



Stack (top at right): J | I | H

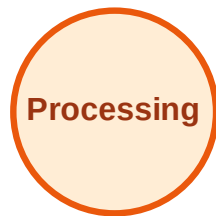
Visited: A, B, D, E, F, G



DFS Traversal

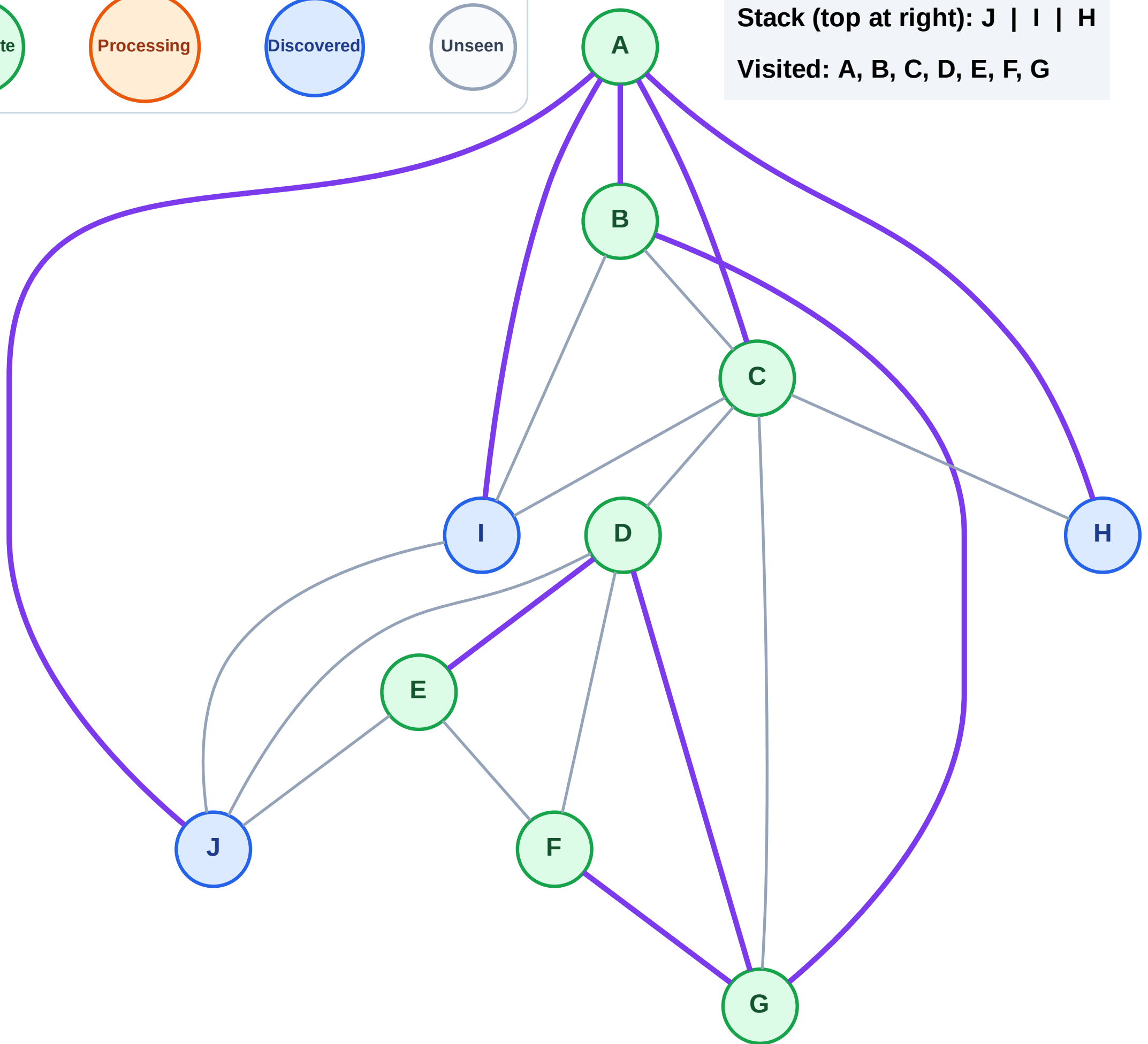
Step 15: No new neighbors from C

Legend



Stack (top at right): J | I | H

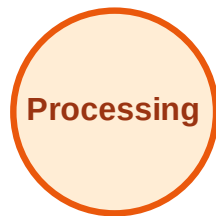
Visited: A, B, C, D, E, F, G



DFS Traversal

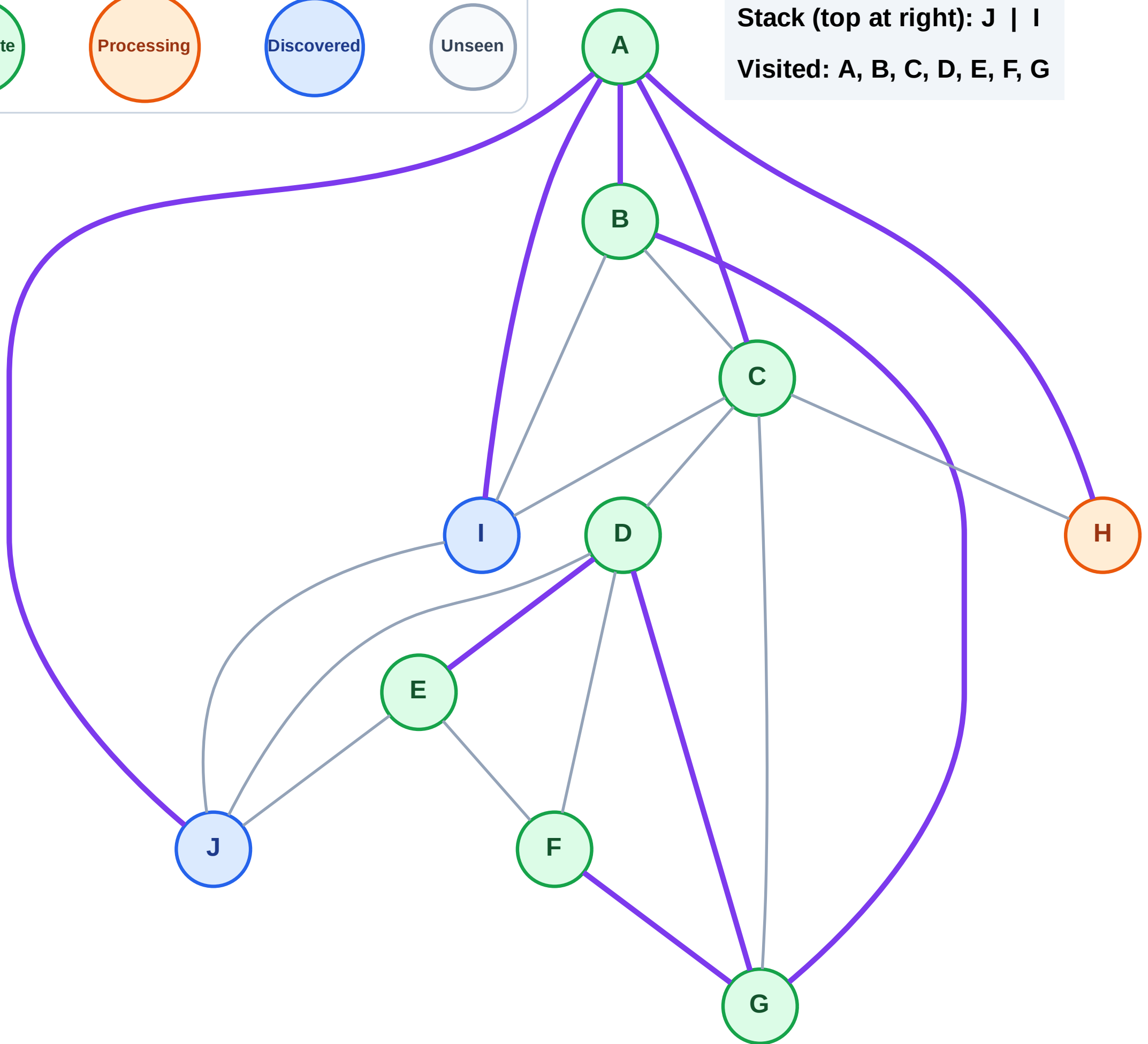
Step 16: Process node H

Legend



Stack (top at right): J | I

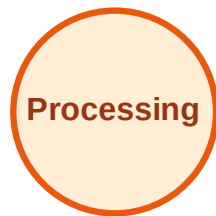
Visited: A, B, C, D, E, F, G



DFS Traversal

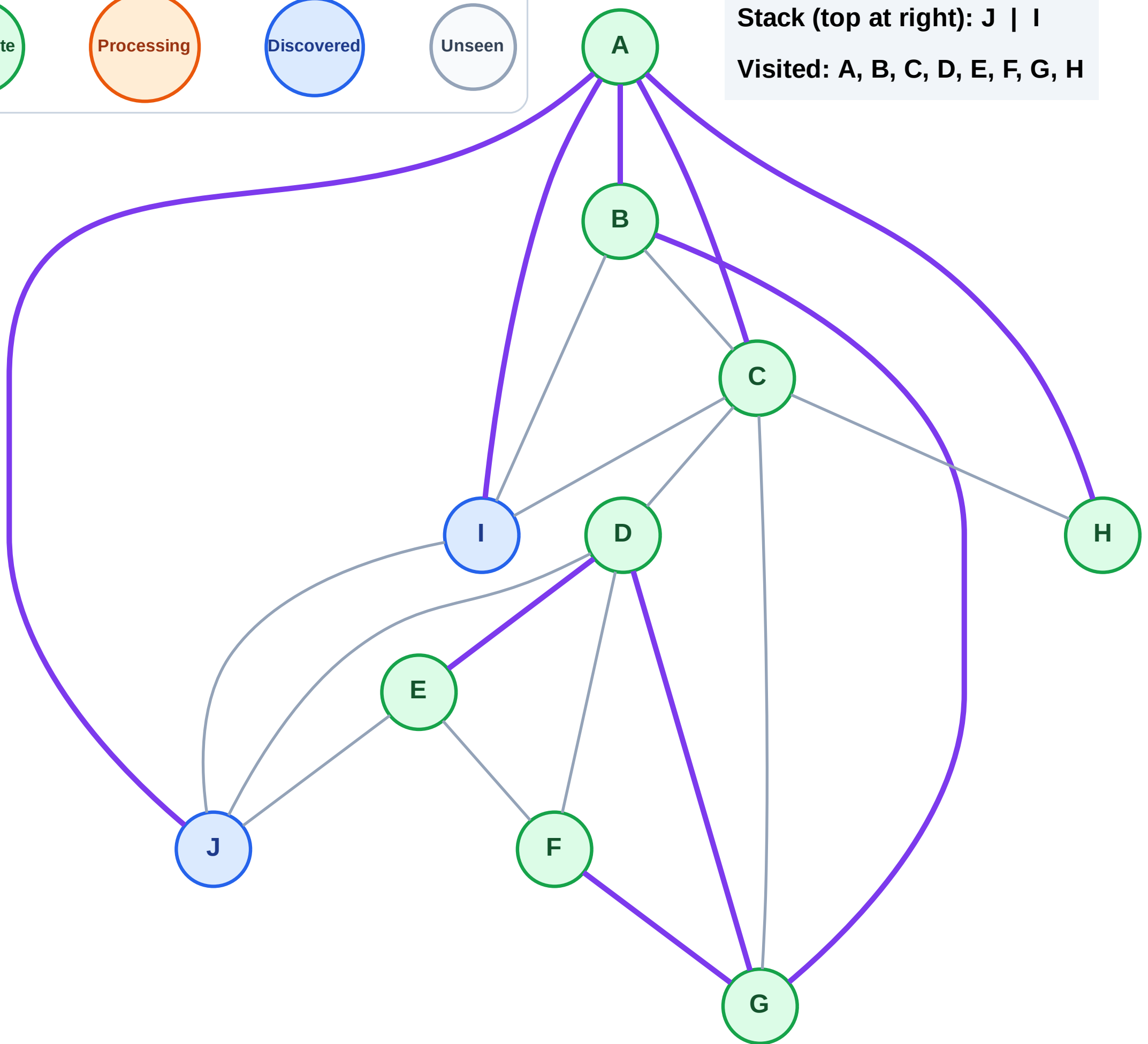
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

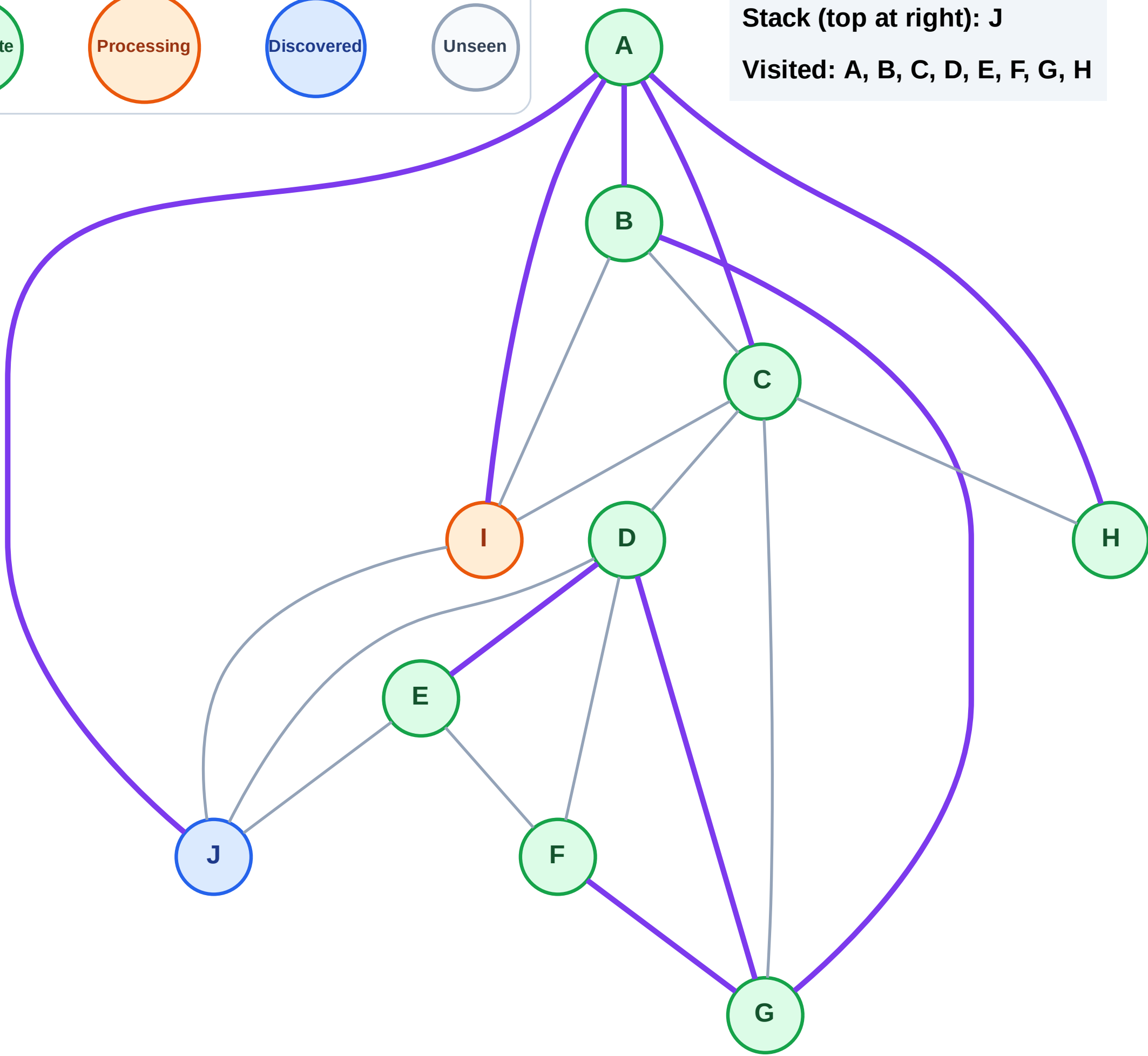
Step 18: Process node I

Legend



Stack (top at right): J

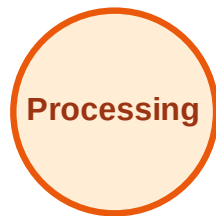
Visited: A, B, C, D, E, F, G, H



DFS Traversal

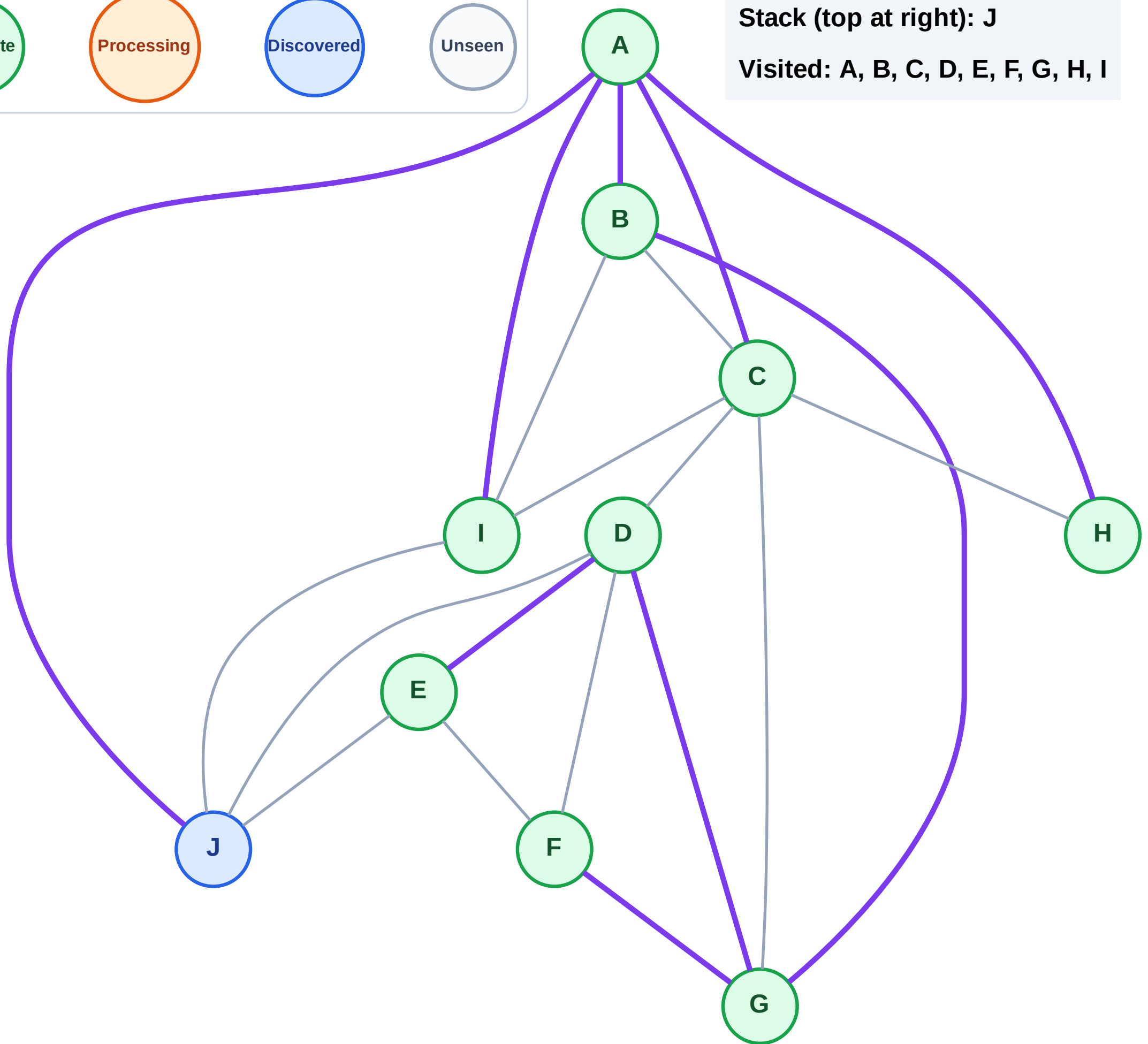
Step 19: No new neighbors from I

Legend



Stack (top at right): J

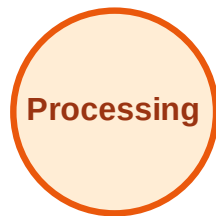
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

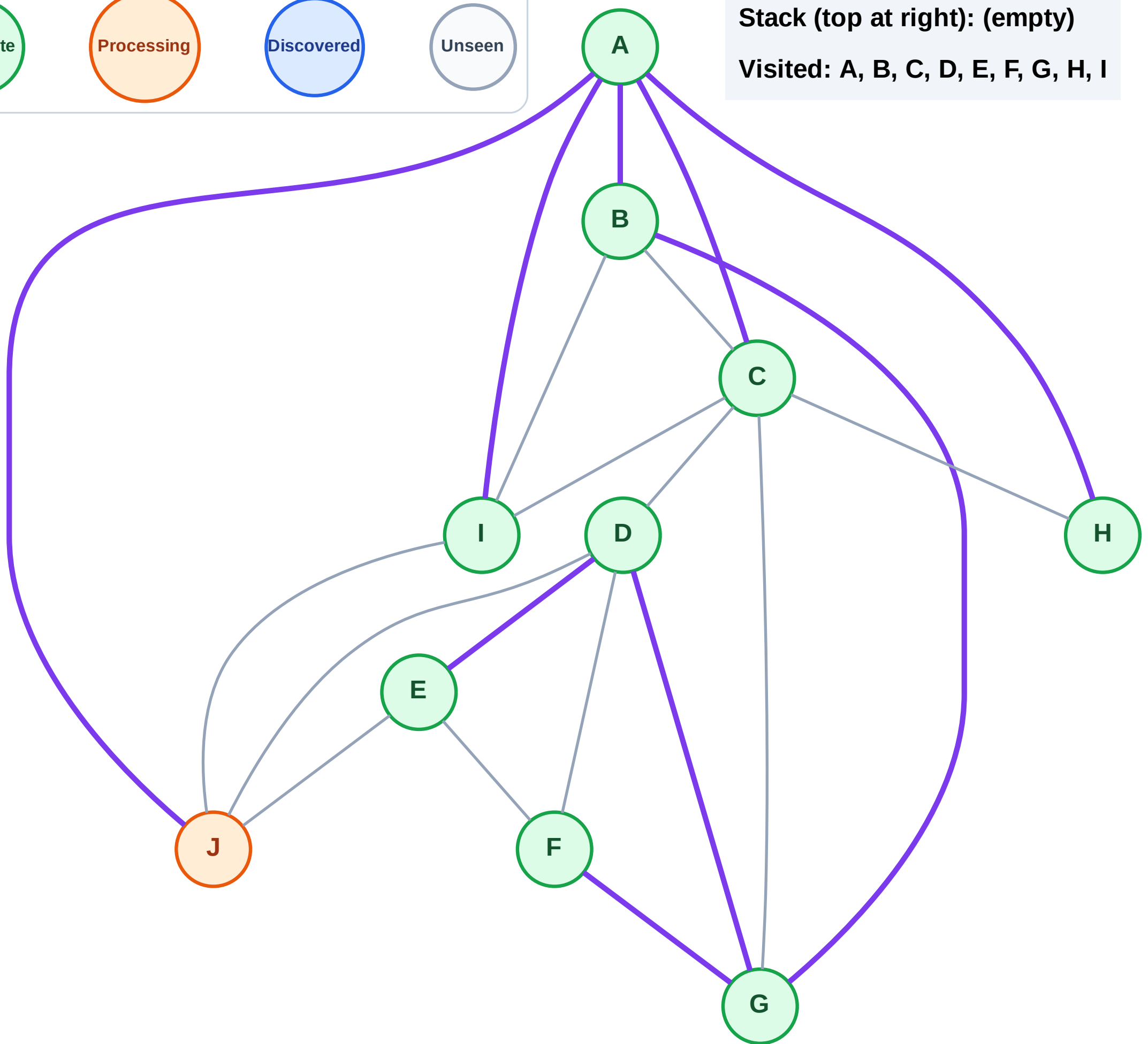
Step 20: Process node J

Legend



Stack (top at right): (empty)

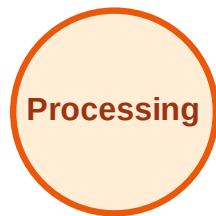
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

